Statistical Foundations

for Quantitative Work

A self-contained tour from data description to modern computational methods

Descriptive statistics · Probability · Random variables
Distributions · Sampling & the CLT · Estimation · Hypothesis testing
Correlation · Linear & multiple regression · Model diagnostics
ANOVA · Nonparametrics · Time series · Logistic regression & GLMs
Bayesian inference · Bootstrap, cross-validation & regularization

A working reference, built to be read slowly and applied.

Preface

This book exists to give you the *statistical* core that everything quantitative rests on: not a grab-bag of formulas, but a connected way of thinking that runs from "here is a column of numbers" all the way to "here is a fitted model and here is why I do or do not trust it." It is deliberately scoped to the foundations. Option theory, key-rate work, and the domain-specific machinery of any particular field are downstream of this material; the goal here is to make that downstream work feel inevitable rather than mysterious.

How this book is organized

The order is not arbitrary. Each part is the prerequisite for the next.

- **Part I — Foundations of Data.** How to summarize and look at data honestly: measures of center and spread, the shape of a distribution, and what skewness and kurtosis actually tell you. Everything later is a principled extension of these basic summaries.
- **Part II — Probability.** The language for uncertainty: probability axioms, conditional probability and Bayes' theorem, random variables, and the specific distributions that recur everywhere. Statistics is applied probability; you cannot reason about a sample without a model for how it was generated.
- **Part III — Inference.** The central magic trick: using a finite sample to make quantified statements about an unknown population. Sampling distributions, the Central Limit Theorem, estimation, confidence intervals, and hypothesis testing.
- **Part IV — Relationships and Regression.** The workhorse of applied quantitative analysis: covariance, correlation, and the full theory of linear and multiple regression, ending with a careful treatment of what goes wrong — heteroskedasticity, autocorrelation, multicollinearity, and influential observations — and how to detect and respond to each.
- **Part V — Further Methods.** A tour of the broader toolkit that builds on the foundations: analysis of variance, nonparametric (rank-based) methods, time-series analysis, logistic regression and the generalized linear model framework, Bayesian inference, and the modern computational methods — bootstrap, cross-validation, and regularization — that define contemporary practice.

How to read it

Read with a pen and a dataset. The chapters are written to be worked through, not skimmed. Three habits will multiply what you get out of them:

1. **Re-derive the short results yourself.** When a variance identity or a least-squares formula is derived in two lines, close the book and reproduce it. The derivations are short on purpose so that this is feasible.
2. **Recompute every numerical example.** Each worked example uses small, deliberately legible numbers so you can check the arithmetic by hand or in a few lines of code, then re-run it on your own data.

3. **Push on the assumptions.** The most valuable parts of statistics are the conditions under which a method fails. Whenever a result is stated, ask what breaks it. Part IV is built almost entirely around that question.

Notation conventions

A few conventions are used throughout. Capital letters such as X, Y denote random variables; lower-case x, y denote particular realized values or fixed constants. A hat denotes an estimator or fitted value: $\hat{\theta}$ estimates the parameter θ, and $\hat{y}_i$ is a fitted value. A bar denotes a sample average, $\bar{x} = \frac{1}{n}\sum_i x_i$. The symbols $\mathbb{E}[\cdot]$, $\mathrm{Var}(\cdot)$, $\mathrm{Cov}(\cdot,\cdot)$, and $\mathrm{Corr}(\cdot,\cdot)$ are expectation, variance, covariance, and correlation. Greek letters generally denote unknown population parameters (the truth we are trying to learn); Latin letters generally denote things we compute from data. Vectors and matrices are written in the natural way and are introduced carefully in Part IV, where they first become essential.

The aim throughout is that nothing is asserted without being explained, and that by the end the machinery feels like common sense with careful bookkeeping attached.

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Part I

Foundations of Data

Chapter 1

Why Statistics?

Statistics is the discipline of reasoning from data under uncertainty. That sentence carries two ideas worth separating. The first is *reasoning from data*: we begin with numbers we have actually observed. The second is *under uncertainty*: those numbers are an incomplete and noisy window onto something larger we genuinely care about but cannot see directly. The entire subject is the tension between these two — what we have versus what we want to know — and the tools for crossing that gap honestly.

1.1 Two kinds of statistics

It is traditional, and genuinely useful, to split the subject in two.

Descriptive statistics summarizes the data in hand. If you have the closing price of an asset for every day last year, computing the average, the spread, and drawing a histogram are descriptive acts. You are not claiming anything beyond the data; you are compressing it into a few interpretable numbers and pictures. Descriptive statistics is, in a sense, just careful arithmetic and good plotting. It is the subject of Part I, and it is foundational precisely because every more sophisticated method is built from these same summaries.

Inferential statistics uses the data in hand to make quantified statements about data *not* in hand. This is the harder and more powerful half. You observe a sample and you want to say something about the larger population it came from, complete with an honest accounting of how wrong you might be. When a poll of two thousand people is used to estimate the opinion of millions, or when last year's returns are used to estimate the volatility you will face next year, that is inference. Parts III and IV are about inference.

The distinction matters because the two activities license different claims. Descriptive statements are, within the data, simply true: the average of these numbers *is* this value. Inferential statements are always probabilistic: based on this sample, the population mean is *plausibly* in this range, and here is the chance we are wrong. Confusing the two — treating an inferential guess as a descriptive certainty — is one of the most common and most expensive errors in applied work.

1.2 Populations and samples

Definition 1.1 (Population and sample)**.** A **population** is the entire collection of units or measurements we wish to understand. A **sample** is the subset of the population that we actually observe.

The population is defined by the question, not by convenience. If you want to understand the behavior of a price spread, the population might be conceived as "all the values that spread could take under the data-generating process," which is infinite and partly hypothetical. If you want to understand customers, the population is the set of all customers, present and perhaps future. The sample is what reality hands you: a finite batch of observations, almost always smaller and messier than you would like.

A *parameter* is a numerical feature of the population — the true mean μ, the true variance σ^2, the true correlation between two quantities. Parameters are fixed but unknown. A *statistic* is a numerical feature computed from the sample — the sample mean $\bar{x}$, the sample variance s^2. Statistics are known (you can compute them) but random (a different sample would give different values). The central project of inference is to use the statistics we can compute to learn about the parameters we cannot. The notational convention reflects this: Greek letters for unknown population parameters, Latin letters for their sample-based estimates.

Example 1.1 (The gap, concretely)**.** Suppose the true average daily value of some quantity is $\mu = 10$, a number you do not know. You collect thirty days of data and compute $\bar{x} = 9.4$. The descriptive fact is exact: the average of *your thirty numbers* is 9.4. The inferential question is what 9.4 tells you about the unknown 10. The answer is never "$\mu = 9.4$." It is a range and a probability: given the spread of the data and the sample size, μ is plausibly somewhere around 9.4 give or take some margin, and the machinery of Part III makes "plausibly" and "some margin" precise.

1.3 Why variability is the whole subject

If every member of a population were identical, statistics would not exist. You would measure one unit and you would be done. Statistics exists because populations vary, and because the act of sampling introduces its own variability on top of that. There are, loosely, two sources of variation to keep distinct.

The first is the *inherent variability of the population itself*: real quantities are dispersed. Prices move, people differ, measurements scatter. This variability is a feature of the world, and a large part of descriptive statistics is measuring it — that is what variance and standard deviation do.

The second is *sampling variability*: because a sample is only part of the population, the statistics you compute from it will differ from the true parameters, and would differ again if you drew a fresh sample. This second kind of variability is what inference must quantify and tame. The remarkable discovery at the heart of the subject — formalized by the Central Limit Theorem in Chapter 6 — is that sampling variability is itself lawful and predictable. We cannot eliminate it, but we can describe it precisely enough to attach honest error bars to our conclusions.

1.4 Types of data

Before summarizing data you must know what kind of data it is, because the kind dictates which summaries and methods are legitimate.

Numerical (quantitative) data are measurements on a meaningful scale. They divide into *discrete* data, which take isolated values, typically counts (the number of trades in an hour, the number of outages in a month), and *continuous* data, which can in principle take any value in a range (a price, a temperature, a duration). Most of this book concerns continuous numerical data, with discrete counts appearing prominently when we study the Bernoulli, binomial, and Poisson distributions in Chapter 5.

Categorical (qualitative) data record group membership rather than magnitude. *Nominal* categories have no natural order (region, asset class, regime label). *Ordinal* categories have an order but no meaningful numerical spacing (a risk rating of low/medium/high). The arithmetic operations that are sensible differ sharply across these types: it is meaningful to average continuous measurements, sometimes meaningful to work with ordinal ranks, and never meaningful to average nominal labels. A surprising amount of bad analysis comes from quietly treating a categorical variable as if it were numerical.

A further distinction that becomes essential in Part IV is between *cross-sectional* data — many units observed at one point in time — and *time-series* data — one unit observed repeatedly over time. Time-series data carry an ordering that cross-sectional data lack, and that ordering creates dependence between nearby observations. This dependence (autocorrelation) violates assumptions that hold comfortably for cross-sectional data and is the source of much of the subtlety in Chapter 12.

1.5 Sampling and the danger of bias

Inference is only as trustworthy as the sample it rests on. The mathematics of Parts III and IV almost always assumes that the sample is, in some sense, *representative* of the population — most cleanly, that observations are drawn at random so that every unit has a known, nonzero chance of selection. When that assumption fails, no amount of sophisticated computation repairs the damage.

Definition 1.2 (Sampling bias)**. Sampling bias** occurs when the procedure used to obtain the sample makes it systematically unrepresentative of the population, so that statistics computed from it are pushed away from the true parameters in a consistent direction.

The dangerous word is *systematic*. Random sampling error is symmetric and self-correcting: average over many samples and it washes out, and we can quantify it. Bias does not wash out. A larger biased sample simply gives you a more precise estimate of the wrong number. The classic forms recur everywhere: *selection bias*, where the available data are not the data you want (only the survivors, only the successes, only the people who responded); *survivorship bias*, a special case where failed units have silently dropped out of the record before you ever see it; and *measurement bias*, where the instrument or definition systematically distorts what is recorded.

The practical discipline this imposes is to interrogate the provenance of your data before you compute anything. Where did these numbers come from? What units *could* have appeared in this dataset but did not, and why? Is anything missing in a way that correlates with what I am trying to measure? These questions sit logically prior to every technique in this book. A beautifully executed regression on a biased sample is a beautifully precise answer to the wrong question.

1.6 The shape of what follows

With this framing in place, the rest of the book has a clear logic. Part I develops the descriptive summaries — center, spread, and shape — that compress a sample into interpretable numbers. Part II builds the probability machinery needed to model how samples are generated, culminating in the catalogue of distributions that recur throughout applied work. Part III is the payoff: the theory that lets a finite sample speak about an unknown population, with quantified uncertainty. Part IV turns to relationships between variables and to regression, the single most used tool in quantitative analysis, and ends by taking seriously every way that tool can mislead you. Throughout, the same two-part definition we began with is the organizing idea: we are always reasoning from data, and always under uncertainty, and the craft lies in doing both honestly at once.

Chapter 2

Describing Data

Given a batch of numbers, the first task is to compress it into a handful of quantities that capture what matters: where the data sit, how spread out they are, and what shape they take. These descriptive summaries are not merely a preliminary nicety. Every inferential method in later chapters is built directly on them — the sample mean and sample variance of this chapter are the raw material of the Central Limit Theorem, confidence intervals, and regression. Getting an honest feel for these summaries, and for how each can mislead, is the foundation of everything else.

Throughout this chapter we work with a sample of n observations $x_1, x_2, \ldots, x_n$.

2.1 Measures of center

A measure of center answers: if I had to name one number as "typical" for this data, what would it be? There are three standard answers, and they disagree in informative ways.

2.1.1 The mean

The **(arithmetic) mean** is the familiar average,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

The mean is the balance point of the data: it is the unique value c for which the deviations sum to zero, $\sum_i (x_i - c) = 0$. It is also the value that minimizes the sum of *squared* deviations, $\sum_i (x_i - c)^2$ — a fact we will meet again as the foundation of least-squares regression in Chapter 10. Because it uses every observation and uses each one's exact magnitude, the mean is efficient and analytically convenient. That same property is its weakness: a single extreme value drags the mean toward itself without limit. The mean of $\{1, 2, 3, 4, 5\}$ is 3; change the last value to 500 and the mean jumps to 102, a "typical" value larger than four of the five observations.

When observations carry different importance, the **weighted mean** generalizes the average:

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i},$$

where w_i is the weight on observation i. Volume-weighted average prices, portfolio returns weighted by position size, and many index calculations are weighted means.

2.1.2 The median

The **median** is the middle value when the data are sorted: the value with half the observations below and half above. For odd n it is the single middle observation; for even n it is conventionally the average of the two middle observations. The median minimizes the sum of *absolute* deviations $\sum_i |x_i - c|$, the counterpart to the mean's squared-deviation property.

The median's defining virtue is **robustness**. Because it depends only on the rank of the middle observation, not on the magnitudes of the extremes, it is essentially immune to outliers. In the example above, the median of $\{1, 2, 3, 4, 5\}$ and of $\{1, 2, 3, 4, 500\}$ is 3 in both cases. Whenever data are skewed or contaminated by extreme values — incomes, house prices, durations, many financial quantities — the median is often the more honest description of the typical case, which is exactly why reported figures for such quantities are usually medians.

2.1.3 The mode

The **mode** is the most frequently occurring value. It is the only measure of center that applies to nominal categorical data (the modal category is simply the most common one), and it is useful for identifying the most likely outcome and for revealing multimodality — the presence of two or more distinct peaks, which often signals that the data are a mixture of qualitatively different groups that should be analyzed separately.

2.1.4 What the comparison tells you

The relationship among the three is diagnostic of shape. In a perfectly symmetric, single-peaked distribution, mean, median, and mode coincide. When they pull apart, the direction of the gap reveals the asymmetry: if the mean exceeds the median, a long right tail of large values is dragging the mean upward; if the mean is below the median, a long left tail is dragging it down. A quick comparison of mean and median is the fastest informal check for skewness, a notion we make precise below.

2.2 Measures of spread

Two datasets can share an identical center and be utterly different because they differ in *dispersion* — how tightly the observations cluster around that center. Spread is, if anything, more important than center for quantitative work, because risk *is* dispersion. A measure of center alone tells you what to expect on average; a measure of spread tells you how badly average can be missed.

2.2.1 Range and interquartile range

The **range**, $\max_i x_i - \min_i x_i$, is the crudest measure: it uses only the two most extreme observations and therefore grows mechanically with sample size and is exquisitely sensitive to outliers. It is occasionally useful as a quick bound but is rarely a serious summary.

Far more useful is the **interquartile range (IQR)**. The *quartiles* divide sorted data into four equal parts: the first quartile Q_1 has one quarter of the data below it, the second quartile Q_2 is the median, and the third quartile Q_3 has three quarters below it. The IQR is their difference,

$$\mathrm{IQR} = Q_3 - Q_1,$$

the width of the middle fifty percent of the data. Because it discards the extreme quarters entirely, the IQR is robust to outliers in the same way the median is, and it is the basis of the common rule that flags an observation as a potential outlier if it lies more than $1.5 \times \mathrm{IQR}$ below Q_1 or above Q_3.

Quantiles (or *percentiles*) generalize this idea: the p-th quantile is the value below which a fraction p of the data falls. The median is the 0.5 quantile, Q_1 the 0.25 quantile, and so on. Quantiles are the natural language for describing the tails of a distribution and reappear throughout the book.

2.2.2 Variance and standard deviation

The dominant measures of spread are built from *deviations* of each observation from the mean, $x_i - \bar{x}$. We cannot simply average these deviations, because by construction they sum to zero. The fix is to remove their signs by squaring before averaging. The **variance** is the average squared deviation.

For an entire population of size N with mean μ, the population variance is

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2.$$

For a *sample*, however, the standard definition divides not by n but by $n-1$:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The replacement of n by $n-1$ is **Bessel's correction**, and the reason for it is worth understanding rather than memorizing. We are using the sample mean $\bar{x}$ as a stand-in for the unknown true mean μ. But $\bar{x}$ is, by its construction, the value that makes $\sum_i (x_i - \bar{x})^2$ as small as possible — smaller than $\sum_i (x_i - \mu)^2$ would be for the true μ. Measuring deviations from the sample's own mean therefore *systematically understates* the true spread. Dividing by the smaller number $n-1$ instead of n inflates the estimate by exactly the right amount to remove this bias on average; one can show that with this divisor $\mathbb{E}[s^2] = \sigma^2$ exactly. The intuition of "degrees of freedom" captures it: once $\bar{x}$ is fixed, only $n-1$ of the deviations are free to vary, because the last is determined by the requirement that they sum to zero. We have spent one degree of freedom estimating the mean, and the divisor reflects it. This idea — that estimating a parameter from the data costs a degree of freedom — recurs constantly, in the t-distribution, the chi-squared distribution, and the denominators of every regression statistic.

Variance has the awkward feature of living in squared units: if the data are in dollars, the variance is in dollars-squared, which is uninterpretable. Taking the square root restores the original units and gives the **standard deviation**,

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

The standard deviation is the single most important measure of spread in the subject. It is, roughly, the typical distance of an observation from the mean, expressed in the data's own units. In finance it is the operational definition of volatility and the basic currency of risk.

A computational identity is frequently useful. Expanding the square gives

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2,$$

so the variance can be computed from the sum of the values and the sum of their squares in a single pass through the data. The same identity, written for random variables as $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, is one of the most-used formulas in all of probability and appears repeatedly in Part II.

Example 2.1 (Variance by hand)**.** Take the five values $4, 8, 6, 5, 7$. Their mean is $\bar{x} = 30/5 = 6$. The deviations are $-2, 2, 0, -1, 1$, with squares $4, 4, 0, 1, 1$ summing to 10. The sample variance is $s^2 = 10/(5-1) = 2.5$, and the sample standard deviation is $s = \sqrt{2.5} \approx 1.58$. So a typical observation sits about 1.58 units from the center of 6. Had we treated these as an entire population, we would have divided by 5 to get $\sigma^2 = 2.0$; the difference between 2.5 and 2.0 is Bessel's correction at work, and it shrinks toward irrelevance as n grows.

2.2.3 Other measures, and comparing spreads across scales

The **median absolute deviation (MAD)**, the median of the absolute deviations from the median, is a fully robust measure of spread, standing to the standard deviation as the median stands to the mean. It is the spread measure of choice when outliers must not be allowed to dominate.

To compare the spread of two datasets measured on different scales — say, a quantity with a mean near 5 against one with a mean near 5000 — absolute spread is misleading, since larger quantities naturally have larger absolute fluctuations. The **coefficient of variation**,

$$\mathrm{CV} = \frac{s}{\bar{x}},$$

expresses the standard deviation as a fraction of the mean, giving a unitless measure of *relative* dispersion that can be compared across scales. (It is only meaningful for data on a ratio scale with a positive mean.)

2.2.4 Standardization and the z-score

Combining a measure of center and a measure of spread lets us put any observation on a common, unit-free footing. The z**-score** of an observation,

$$z_i = \frac{x_i - \bar{x}}{s},$$

records how many standard deviations the observation lies above (if positive) or below (if negative) the mean. Standardization strips away the original units and scale, so that values from different variables become directly comparable, and it is the device that lets a single tabulated standard normal distribution serve for all normal distributions in Chapter 5. A z-score of $+2$ always means "two standard deviations above average," whether the underlying quantity is a temperature, a price, or a test score.

2.3 The shape of a distribution

Center and spread are the first two summaries; shape is the third. Two distributions can agree on mean and standard deviation yet differ profoundly in their asymmetry and in the heaviness of their tails. These features are captured by higher *moments* of the data, and for risk-bearing quantities they are not academic refinements — they are often where the danger lives.

2.3.1 Skewness

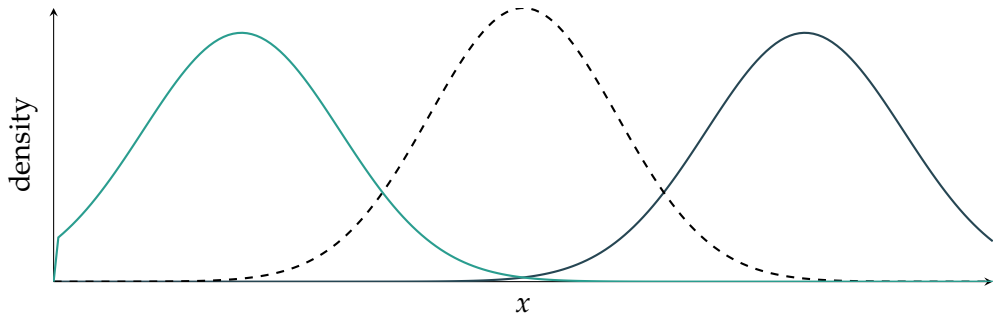
Skewness measures asymmetry. It is built from the average *cubed* standardized deviation:

$$g_1 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3.$$

Cubing preserves sign, so deviations above and below the mean do not cancel; instead, large deviations are amplified, and the side with the longer, heavier tail dominates the sum. A distribution with a long right tail (a few very large values) has *positive* skew; one with a long left tail has *negative* skew; a symmetric distribution has zero skew. As noted earlier, positive skew typically pulls the mean above the median, and negative skew pulls it below, which is why the mean–median gap is a quick informal read on skewness.

Skewness matters enormously in practice because many real quantities are inherently asymmetric. Prices and many financial returns, durations, and quantities bounded below by zero but unbounded above are commonly right-skewed. When a quantity is skewed, a symmetric summary like "μ plus or minus a standard deviation" misrepresents the risk, understating exposure on the heavy-tailed side.

Left-skewed, symmetric, and right-skewed shapes



2.3.2 Kurtosis and fat tails

Kurtosis measures the heaviness of the tails relative to a normal distribution. It is built from the average *fourth* standardized deviation:

$$g_2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4.$$

The fourth power weights extreme deviations even more aggressively than the third, so kurtosis is dominated by the rare, very large excursions in the tails. For the normal distribution this quantity equals 3, so it is conventional to report **excess kurtosis**, $g_2 - 3$, which is zero for a normal distribution, positive for a distribution with heavier tails and a sharper central peak (*leptokurtic*), and negative for one with lighter tails (*platykurtic*).

Of all the shape diagnostics, excess kurtosis is arguably the most consequential for quantitative finance, because it is the formal statement that extreme events happen more often than a normal model predicts. Empirical return distributions are notoriously leptokurtic: the large moves that a normal distribution treats as essentially impossible occur with uncomfortable regularity. A risk model that assumes normality — that assumes zero excess kurtosis — will systematically and dangerously underestimate the probability of large losses. We will return to this point sharply in Chapter 5, where the lognormal and heavy-tailed distributions are introduced, and again in Chapter 12, where non-normal, heavy-tailed residuals undermine the standard inferential machinery of regression.

2.4 Visualizing distributions

Numerical summaries compress, and compression discards. Two datasets can share mean, variance, skewness, and kurtosis and still look completely different. Plotting the data is therefore not optional; it is the only way to see structure that summaries hide — multimodality, gaps, clusters, and outliers. Four plots do most of the work.

2.4.1 The histogram

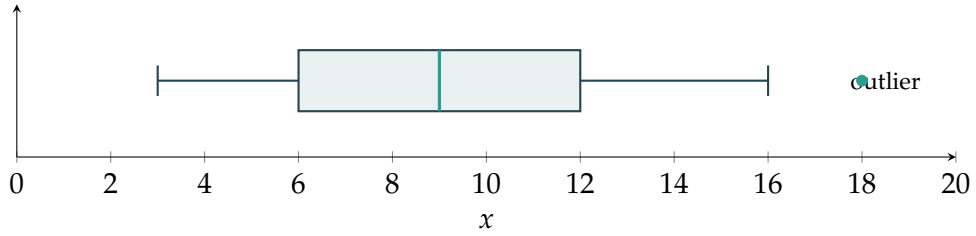
A **histogram** partitions the range of the data into adjacent intervals (*bins*) and draws a bar over each whose height is the count (or proportion) of observations falling in it. It is the most direct

picture of a distribution's shape, immediately revealing center, spread, skewness, the number of peaks, and obvious outliers. Its one genuine subtlety is the choice of bin width: too few bins over-smooth and hide structure, while too many produce a jagged picture dominated by sampling noise. The bin width is a smoothing parameter, and the honest practice is to look at the data under more than one choice. A histogram normalized so that the total area equals one is an empirical estimate of the underlying probability density of Chapter 4.

2.4.2 The boxplot

A **boxplot** (box-and-whisker plot) is a compact summary built entirely from quantiles. A box spans from Q_1 to Q_3, with a line at the median; whiskers extend to the most extreme observations within $1.5 \times$ IQR of the box, and points beyond the whiskers are drawn individually as candidate outliers. The boxplot sacrifices the fine detail of the histogram — it cannot show multimodality, since it is built only from quantiles — but it excels at displaying the median, the spread of the central half, the skewness (visible as an off-center median or unequal whisker lengths), and the outliers, all at a glance. Its real strength is comparison: several boxplots side by side make differences in center and spread across groups immediately legible.

Boxplot: box spans Q_1 to Q_3, line at median, whiskers to 1.5×IQR



2.4.3 The empirical CDF

The **empirical cumulative distribution function (ECDF)** plots, for each value x, the fraction of the data less than or equal to x:

$$\hat{F}(x) = \frac{\#\{i : x_i \leq x\}}{n}.$$

It is a non-decreasing step function rising from 0 to 1, jumping by $1/n$ at each observation. Unlike the histogram it requires no binning choice and so involves no arbitrary smoothing; it displays the data exactly. It is the natural plot for reading off quantiles directly (find the height p and read across to the corresponding value), and it is the sample analogue of the cumulative distribution function that is central to Chapter 4. It is also the right tool for comparing two distributions, since two ECDFs overlaid reveal precisely where and how the distributions differ.

2.4.4 The Q–Q plot

A **quantile–quantile (Q–Q) plot** compares the shape of a dataset to a reference distribution — usually the normal — by plotting the sorted data values against the values you would expect if the data truly came from that reference. If the data follow the reference distribution, the points fall on a straight line. Systematic departures from the line are highly informative and have characteristic signatures: an S-shape indicates skewness, while points that curve away from the line at both ends — falling below on the left and above on the right — indicate *heavy tails*, the visual fingerprint of the positive excess kurtosis discussed above. The Q–Q

plot is the standard graphical check for normality, and it is the diagnostic we will deploy in Chapter 12 to assess whether regression residuals are plausibly normal, a question on which several inferential procedures depend.

2.5 Summary

Describing data means reporting three things honestly: a center, a spread, and a shape. The mean and median offer complementary notions of center, trading efficiency against robustness; the standard deviation and IQR do the same for spread; and skewness and kurtosis quantify the asymmetry and tail-heaviness that a center and spread alone cannot convey. None of these numbers substitutes for looking: the histogram, boxplot, ECDF, and Q–Q plot each reveal structure the summaries discard, and the heavy tails that they expose are, for risk-bearing quantities, frequently the most important feature of all. Every inferential method in the chapters that follow is assembled from exactly these summaries — above all from the sample mean and sample variance — so the intuition built here is the foundation on which the rest of the book stands.

Part II

Probability

Chapter 3

Probability

Descriptive statistics summarizes data we have. To reason from that data to conclusions about data we do not have, we need a mathematical language for uncertainty itself. That language is probability. Probability is the theory of how likelihoods behave — how they combine, how they update in light of new information, and how they constrain one another. Every inferential method in this book is, underneath, an application of probability, so this chapter and the two that follow build the machinery carefully from the ground up.

3.1 Sample spaces and events

We begin with a conceptual experiment whose outcome is uncertain.

Definition 3.1 (Sample space and events)**.** The **sample space** Ω is the set of all possible outcomes of an experiment. An **event** is any subset $A \subseteq \Omega$ — a collection of outcomes. We say the event A *occurs* if the realized outcome lies in A.

If the experiment is a single roll of a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$, and "the roll is even" is the event $A = \{2, 4, 6\}$. If the experiment is observing tomorrow's value of a continuous quantity, Ω might be the entire real line, and "the value exceeds 100" is the event $(100, \infty)$. Because events are sets, the operations of set theory are exactly the operations we need for combining events: the *union* $A \cup B$ is the event that A or B (or both) occurs; the *intersection* $A \cap B$ is the event that both occur; and the *complement* A^c is the event that A does not occur. Two events are *mutually exclusive* (or *disjoint*) if they cannot occur together, $A \cap B = \varnothing$.

3.2 The axioms of probability

Probability assigns to each event a number measuring how likely it is. Rather than define what that number "really means," modern probability proceeds axiomatically: it lays down a small set of rules that any sensible assignment must obey, due to Kolmogorov.

Definition 3.2 (Probability axioms)**.** A **probability** is a function $\mathbb{P}$ assigning to each event A a real number $\mathbb{P}(A)$ such that

(i) $\mathbb{P}(A) \geq 0$ for every event A (non-negativity);

(ii) $\mathbb{P}(\Omega) = 1$ (the certain event has probability one);

(iii) if $A_1, A_2, \ldots$ are mutually exclusive, then $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$ (additivity).

Everything else in probability is a logical consequence of these three rules. A few consequences are used constantly. From the axioms one derives the **complement rule**,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A),$$

which follows because A and A^c are disjoint and together fill Ω. One also obtains **monotonicity** — if $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$ — and the bound $0 \leq \mathbb{P}(A) \leq 1$ for every event. For events

that are *not* disjoint, simple addition double-counts the overlap, and the correction is the **inclusion–exclusion** formula:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

The subtracted term removes the outcomes in both A and B, which the first two terms each counted once.

3.2.1 Equally likely outcomes and counting

When a sample space is finite and symmetry makes every outcome equally likely, probability reduces to counting:

$$\mathbb{P}(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega}.$$

Evaluating such probabilities requires counting the size of sets, the province of *combinatorics*. Two tools suffice for most purposes. The number of ways to arrange k items from n distinct items in order is the number of *permutations*, $n!/(n-k)!$. The number of ways to choose k items from n *without regard to order* is the number of *combinations*,

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!},$$

read "n choose k." The binomial coefficient $\binom{n}{k}$ will reappear in Chapter 5 as the count of the ways a given number of successes can occur, which is exactly what gives the binomial distribution its shape.

3.3 Conditional probability

The most important idea in this chapter is that probabilities change as information arrives. The probability that a quantity exceeds some threshold is one thing in the abstract and quite another once you have learned that a related quantity is unusually high. Conditional probability formalizes this updating.

Definition 3.3 (Conditional probability)**.** The **conditional probability** of A given B, for an event B with $\mathbb{P}(B) > 0$, is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The definition is more intuitive than it first appears. Learning that B has occurred restricts the world to the outcomes inside B; that set becomes the new, smaller sample space. Within that restricted world, the chance of A is the share of B that also belongs to A — hence the ratio of $\mathbb{P}(A \cap B)$ to $\mathbb{P}(B)$. The denominator renormalizes so that, conditional on B, the probabilities again sum to one.

Rearranging the definition gives the **multiplication rule**, a workhorse for computing the probability of joint occurrences:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B) = \mathbb{P}(B \mid A)\,\mathbb{P}(A).$$

The probability that both events happen is the probability of one, times the probability of the other given the first.

3.4 Independence

Sometimes learning that B occurred tells you nothing about A. This is the precise meaning of independence.

Definition 3.4 (Independence)**.** Events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B),$$

equivalently (when the conditionals are defined) if $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

The equivalent form is the more revealing one: A and B are independent exactly when conditioning on B leaves the probability of A unchanged. Independence is a strong and frequently *assumed* property, and it is one of the most consequential assumptions in all of applied statistics, because it is what licenses multiplying probabilities and, more deeply, what underlies the standard error formulas and the Central Limit Theorem of Chapter 6. It is also frequently false in ways that matter. Two warnings recur throughout the book. First, independence must not be confused with mutual exclusivity: disjoint events with positive probability are in fact strongly *dependent*, since knowing one occurred tells you the other did not. Second, observations spaced through time — the time-series data of Chapter 12 — are routinely *not* independent, because nearby values are correlated; assuming independence where it does not hold is a classic and costly error that we will study at length under the name autocorrelation.

3.5 The law of total probability

Often the probability of an event is hard to compute directly but easy to compute within each of several distinct scenarios. The law of total probability stitches the scenarios back together.

Definition 3.5 (Partition)**.** Events $B_1, \dots, B_k$ form a **partition** of Ω if they are mutually exclusive and together exhaust the sample space ($\bigcup_i B_i = \Omega$). Exactly one of them occurs.

Theorem 3.1 (Law of total probability)**.** *If $B_1, \dots, B_k$ partition Ω, then for any event A,*

$$\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i).$$

The formula computes the overall probability of A as a weighted average of its conditional probabilities across the scenarios, each weighted by how likely that scenario is. If a quantity behaves one way in a high-volatility regime and another way in a calm regime, the unconditional probability of an event is the regime-conditional probability in each, weighted by the probability of being in that regime. This decomposition — average the conditional answers, weighting by the probability of each condition — is one of the most useful problem-solving moves in the subject, and it is precisely the engine of Bayes' theorem.

3.6 Bayes' theorem

Conditional probability is not symmetric: $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ are different quantities, and confusing them is the single most common probabilistic fallacy. Bayes' theorem is the rule that correctly converts one into the other, and it is the formal foundation of learning from evidence.

Theorem 3.2 (Bayes' theorem)**.** *For events A and B with positive probability,*

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\,\mathbb{P}(B)}{\mathbb{P}(A)}.$$

If $B_1, \dots, B_k$ partition Ω, the denominator may be expanded by the law of total probability:

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)}{\sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}.$$

The theorem follows immediately from writing the multiplication rule two ways and equating, but its interpretation is what makes it profound. Read B as a hypothesis and A as observed evidence. The term $\mathbb{P}(B)$ is the *prior* probability of the hypothesis before seeing the evidence; $\mathbb{P}(A \mid B)$ is the *likelihood*, the probability the evidence would arise if the hypothesis were true; and $\mathbb{P}(B \mid A)$ is the *posterior*, the updated probability of the hypothesis after the evidence is in. Bayes' theorem is therefore the mathematics of rational belief revision: it tells you exactly how much to update your view of a hypothesis when you observe data that bears on it.

Example 3.1 (Why base rates dominate: the rare-condition test)**.** This classic example is worth working in full, because the lesson it teaches recurs whenever a rare event is being detected. Suppose a condition is present in 1% of a population, so the prior is $\mathbb{P}(\text{present}) = 0.01$. A test for it is fairly accurate: if the condition is present, the test is positive 99% of the time, $\mathbb{P}(+ \mid \text{present}) = 0.99$; if absent, it is wrongly positive only 5% of the time, $\mathbb{P}(+ \mid \text{absent}) = 0.05$. A randomly chosen individual tests positive. What is the probability the condition is actually present?

The intuitive guess is "about 99%, since the test is 99% accurate." Bayes' theorem shows this is badly wrong. The denominator, the overall probability of a positive test, is by the law of total probability

$$\mathbb{P}(+) = \underbrace{0.99 \times 0.01}_{\text{true positives}} + \underbrace{0.05 \times 0.99}_{\text{false positives}} = 0.0099 + 0.0495 = 0.0594.$$

Therefore

$$\mathbb{P}(\text{present} \mid +) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} \approx 0.167.$$

Despite the positive result on an apparently accurate test, the condition is present only about one time in six. The reason is the base rate: the condition is so rare that the small false-positive rate, applied to the vast majority who do not have it, generates more false positives than the test generates true positives among the few who do. The prior dominates. The general lesson — that the reliability of a signal cannot be read off its accuracy alone but depends critically on the base rate of what it is detecting — is one of the most important and most frequently ignored ideas in all of applied probability, and it has obvious force whenever one is trying to detect rare, high-impact events against a noisy background.

3.7 Summary

Probability is built from three axioms, and everything else is consequence. The complement, inclusion–exclusion, and the bound on probabilities follow at once. Conditional probability formalizes how likelihoods change with information; independence is the special case in which they do not change, and is a powerful but easily misused assumption. The law of total probability decomposes a hard probability into easy conditional pieces, and Bayes' theorem

inverts conditional probabilities to give the mathematics of learning from evidence — with the standing warning, dramatized by the rare-condition test, that base rates govern how much a signal is really worth. With this language for uncertainty in place, the next chapter attaches numbers to outcomes and studies random variables, the bridge from probability to the quantitative summaries of Part I.

Chapter 4

Random Variables

Probability as developed so far attaches numbers to events. In quantitative work we want to attach numbers to *outcomes* — a price, a return, a count, a duration — and then study the behavior of those numbers. A random variable is the device that does this. It is the central object of probability and the bridge to the descriptive summaries of Part I: the mean and variance of a random variable are the population analogues of the sample mean and sample variance of Chapter 2, and the whole enterprise of inference is the attempt to learn the former from the latter.

4.1 Random variables and their distributions

Definition 4.1 (Random variable)**.** A **random variable** X is a function that assigns a real number to each outcome in the sample space. It is **discrete** if it takes values in a countable set (often integer counts), and **continuous** if it can take any value in an interval of the real line.

The distinction is not pedantry; discrete and continuous variables are described by different objects and require different operations (sums versus integrals). We treat each in turn and then unify them through the cumulative distribution function.

4.1.1 Discrete variables: the probability mass function

A discrete random variable is fully described by listing the probability of each value it can take.

Definition 4.2 (Probability mass function)**.** The **probability mass function (PMF)** of a discrete random variable X is

$$p(x) = \mathbb{P}(X = x),$$

satisfying $p(x) \geq 0$ for all x and $\sum_x p(x) = 1$.

The PMF assigns to each possible value the probability that the variable equals exactly that value. The two conditions are simply the probability axioms in this setting: probabilities are non-negative, and because the variable must take *some* value, the probabilities over all values sum to one. The probability that X falls in a set is obtained by summing the mass over the values in that set.

4.1.2 Continuous variables: the probability density function

For a continuous variable, the probability of any single exact value is zero — there are uncountably many possible values, so no individual one can carry positive probability. Probability instead attaches to *ranges*, and is described by a density.

Definition 4.3 (Probability density function)**.** A continuous random variable X has a **probability density function (PDF)** $f(x) \geq 0$ such that the probability of landing in an interval is the area under f over that interval:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

The crucial conceptual point is that $f(x)$ is *not* a probability — it is a probability *density*, a probability per unit length, and it may exceed one. Only when integrated over an interval does it yield a probability. The total area under the density is one, the continuous echo of the requirement that the PMF sums to one. A normalized histogram from Chapter 2 is precisely an empirical estimate of this density.

4.1.3 The cumulative distribution function

There is a single object that describes *both* discrete and continuous variables uniformly, and that is so useful it deserves to be regarded as the fundamental descriptor of a distribution: the cumulative distribution function.

Definition 4.4 (Cumulative distribution function)**.** The **cumulative distribution function (CDF)** of any random variable X is

$$F(x) = \mathbb{P}(X \leq x).$$

The CDF accumulates probability from the left: $F(x)$ is the total probability that the variable comes out at or below the value x. It is defined identically for discrete and continuous variables, which is its great unifying virtue. Every CDF shares three properties that follow directly from the axioms: it is non-decreasing (accumulating more can only increase the total), it runs from 0 at the far left to 1 at the far right ($F(-\infty) = 0$, $F(+\infty) = 1$), and probabilities of intervals are read off as differences,

$$\mathbb{P}(a < X \leq b) = F(b) - F(a).$$

For a discrete variable the CDF is a step function, jumping at each value by that value's mass. For a continuous variable the CDF is smooth, and it is tied to the density by the fundamental theorem of calculus: the CDF is the integral of the density, and conversely the density is the derivative of the CDF,

$$F(x) = \int_{-\infty}^{x} f(t)\,dt, \qquad f(x) = F'(x).$$

This relationship is worth dwelling on, because the CDF is the workhorse of applied probability. The quantiles of Chapter 2 are defined through it (the p-th quantile is the value x with $F(x) = p$, i.e. the inverse CDF evaluated at p); the empirical CDF of Chapter 2 is its sample estimate; and, most importantly for applied work, the CDF is exactly the tool that converts a model of a quantity into the probability of a threshold event. If you have a distribution for some future quantity and you want the probability that it lands above a level c, that probability is $1 - F(c)$ — a direct evaluation of the CDF. Whenever a forecast distribution must be turned into the probability of clearing or missing a threshold, it is the CDF doing the work.

4.2 Expectation

A distribution is a complete description; often we want a single number summarizing its center. That number is the expectation, the population counterpart of the sample mean.

Definition 4.5 (Expectation)**.** The **expectation** (or **mean**) of a random variable X is its probability-weighted average value:

$$\mathbb{E}[X] = \sum_{x} x\,p(x) \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int_{-\infty}^{\infty} x\,f(x)\,dx \quad \text{(continuous)}.$$

The expectation is the long-run average value the variable would produce over many independent repetitions — the balance point of the distribution, exactly as the sample mean is the balance point of a dataset. We often write $\mu = \mathbb{E}[X]$.

The single most useful property of expectation is **linearity**, which holds with no conditions whatsoever:

$$\mathbb{E}[aX + bY + c] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y] + c.$$

The expectation of a sum is the sum of the expectations, and constants pull through. Remarkably, this holds even when X and Y are dependent — no independence assumption is needed. Linearity of expectation is one of the most powerful labor-saving devices in all of probability; many results that look forbidding when approached head-on collapse immediately once expectation is distributed across a sum.

It is essential to note what linearity does *not* extend to: in general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$ for a nonlinear function g. The expectation of a square is not the square of the expectation; the expectation of a ratio is not the ratio of expectations. To compute the expectation of a transformed variable, one uses the rule sometimes called the law of the unconscious statistician,

$$\mathbb{E}[g(X)] = \sum_x g(x)\,p(x) \qquad \text{or} \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\,f(x)\,dx,$$

which weights the *transformed* values by the original probabilities. The gap between $\mathbb{E}[g(X)]$ and $g(\mathbb{E}[X])$ is not a technicality; for convex g it has a definite sign (by Jensen's inequality, $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$ when g is convex), and this gap is the mathematical origin of several important phenomena, including the difference between the arithmetic and geometric behavior of compounded returns.

4.3 Variance and standard deviation

The expectation locates the center; the variance measures spread around it, mirroring exactly the development of Chapter 2 but now for the underlying distribution rather than a sample.

Definition 4.6 (Variance and standard deviation)**.** The **variance** of X with mean μ is the expected squared deviation from the mean,

$$\operatorname{Var}(X) = \mathbb{E}\left[(X - \mu)^2\right],$$

and the **standard deviation** is $\operatorname{SD}(X) = \sqrt{\operatorname{Var}(X)}$, often written σ.

Expanding the square and using linearity yields the indispensable computational identity, the population version of the one met in Chapter 2:

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The variance is the mean of the square minus the square of the mean. This identity is used constantly, and it immediately implies the useful fact that $\mathbb{E}[X^2] = \operatorname{Var}(X) + (\mathbb{E}[X])^2 \geq (\mathbb{E}[X])^2$.

The variance behaves in a specific and important way under linear transformations. For constants a and b,

$$\operatorname{Var}(aX + b) = a^2\,\operatorname{Var}(X).$$

Two features deserve emphasis. First, adding a constant b shifts the whole distribution but does not change its spread, so b disappears. Second, scaling by a multiplies the variance by a^2, not by a — because variance lives in squared units. In terms of standard deviation this reads $\operatorname{SD}(aX + b) = |a|\,\operatorname{SD}(X)$, so the standard deviation scales linearly with a, which is one reason it, rather than the variance, is the natural unit for risk.

4.3.1 Moments and the moment generating function

The mean and variance are the first two of an infinite sequence of summaries called *moments*. The k-th moment is $\mathbb{E}[X^k]$, and the k-th *central* moment is $\mathbb{E}[(X-\mu)^k]$. The first moment is the mean; the second central moment is the variance; the third and fourth central moments, suitably standardized, are the skewness and kurtosis of Chapter 2, now expressed for the distribution itself. The moments collectively encode the shape of a distribution in increasing detail.

These moments can be packaged into a single function. The **moment generating function (MGF)** is

$$M_X(t) = \mathbb{E}\left[e^{tX}\right],$$

when this expectation exists for t near zero. Its name comes from the fact that differentiating it k times and evaluating at $t = 0$ produces the k-th moment, $M_X^{(k)}(0) = \mathbb{E}[X^k]$. The MGF has two properties that make it genuinely useful rather than merely a curiosity. First, it uniquely determines the distribution: if two variables have the same MGF, they have the same distribution. Second, the MGF of a sum of *independent* variables is the product of their MGFs, which turns the difficult problem of finding the distribution of a sum into the easy problem of multiplying functions. This second property is the cleanest route to several results in Chapter 5 — for instance, that a sum of independent normal variables is again normal — and it underlies one standard proof of the Central Limit Theorem.

4.4 Joint distributions and dependence

Quantitative problems almost always involve several variables at once, and the questions that matter most concern how they move *together*. This requires describing variables jointly.

For two discrete variables the **joint PMF** is $p(x,y) = \mathbb{P}(X = x, Y = y)$; for two continuous variables the **joint PDF** $f(x,y)$ gives probabilities as volumes, $\mathbb{P}((X,Y) \in A) = \iint_A f(x,y)\,dx\,dy$. From a joint distribution we recover the behavior of a single variable by summing or integrating out the other, producing the **marginal** distribution — for example $p_X(x) = \sum_y p(x,y)$. And we describe one variable given knowledge of the other through the **conditional** distribution, the direct analogue of conditional probability:

$$p_{Y|X}(y \mid x) = \frac{p(x,y)}{p_X(x)}.$$

Definition 4.7 (Independence of random variables)**.** Random variables X and Y are **independent** if their joint distribution factors into the product of their marginals,

$$p(x,y) = p_X(x)\,p_Y(y) \quad \text{(discrete)}, \qquad f(x,y) = f_X(x)\,f_Y(y) \quad \text{(continuous)},$$

for all x, y. Equivalently, the conditional distribution of Y given X does not depend on X.

Independence of random variables is the precise condition under which one variable carries no information about the other, and it is the assumption that underlies the simplest and most-used results in the rest of the book. When X and Y are independent, expectations of products factor, $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$, a fact we will use immediately below and which is the gateway to covariance in Chapter 9.

4.5 Sums of random variables

The behavior of *sums* of random variables is so central to statistics that it deserves separate emphasis. Inference is built on the sample mean, which is a scaled sum; the standard error

formulas of Part III are statements about the variance of a sum; and the Central Limit Theorem of Chapter 6 is a statement about the limiting distribution of a sum. Two results carry most of the weight.

For the *expectation* of a sum, linearity gives the answer unconditionally:

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

No assumptions are required. The *variance* of a sum is more delicate, because it depends on how the variables co-move. The general formula is

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j),$$

where the covariance $\mathrm{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]$ measures co-movement and is studied fully in Chapter 9. The cross terms capture the fact that when variables move together, their fluctuations reinforce, inflating the variance of the total; when they move oppositely, fluctuations partially cancel, reducing it. This single formula is the mathematical heart of diversification: the risk of a sum of positions is not the sum of their risks unless the positions are uncorrelated.

In the important special case where the variables are *independent* (or merely uncorrelated), every covariance term vanishes and the formula collapses to the clean statement that *variances add*:

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) \qquad \text{(independent case)}.$$

From this one fact the most important formula in inferential statistics follows in a single line. Suppose $X_1, \ldots, X_n$ are independent with common mean μ and common variance σ^2. The sample mean is $\bar{X} = \frac{1}{n}\sum_i X_i$. Its expectation, by linearity, is $\mathbb{E}[\bar{X}] = \mu$ — the sample mean is centered on the true mean. Its variance, using $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$ with $a = 1/n$ together with the additivity of variance, is

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

Taking the square root gives the **standard error** of the mean, $\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$. This result is the engine of inference. It says that averaging reduces variance: the spread of the sample mean shrinks as the sample grows, and it shrinks in proportion to $1/\sqrt{n}$. Quadrupling the sample halves the uncertainty in the mean. The entire apparatus of confidence intervals and hypothesis tests in Part III is built on this $\sigma/\sqrt{n}$, and it is worth recognizing that it rests on exactly two ingredients developed here — the linearity of expectation and the additivity of variance under independence.

4.6 Summary

A random variable attaches numbers to outcomes, and its distribution is described by a PMF (discrete) or PDF (continuous), unified by the CDF, which accumulates probability from the left and is the natural tool for turning a distribution into the probability of a threshold event. The expectation is the probability-weighted center and obeys the powerful, assumption-free property of linearity; the variance measures squared spread, scales as a^2 under multiplication, and is computed through $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$. Higher moments encode shape, and the MGF

packages them all while taming sums of independent variables. Joint distributions describe several variables together, with independence the condition of no shared information. Above all, the behavior of sums — expectations always add, variances add when independence holds — yields the standard error $\sigma/\sqrt{n}$, the single result on which Part III is built. The next chapter populates this framework with the specific distributions that recur throughout applied quantitative work.

Chapter 5

The Distributions That Matter

A distribution is a model for how a random quantity behaves. A relatively small number of distributions arise again and again, because each corresponds to a canonical mechanism — a yes/no trial, a count of rare events, a sum of many small influences, a multiplicative process. Knowing this catalogue, and the mechanisms behind each member, lets you recognize the right model for a situation and import everything that is known about it. This chapter is that catalogue. For each distribution we give its form, its mean and variance, the mechanism it models, and its relationships to the others, because the distributions are not isolated facts but a connected family.

5.1 Discrete distributions

5.1.1 The Bernoulli distribution: the atom

The simplest non-trivial random variable encodes a single yes/no trial.

Definition 5.1 (Bernoulli)**.** A **Bernoulli** random variable $X \sim \text{Bernoulli}(p)$ takes the value 1 ("success") with probability p and 0 ("failure") with probability $1 - p$. Its PMF is $p(1) = p$, $p(0) = 1 - p$.

Direct computation from the definition of expectation gives the mean and variance, and they are worth committing to memory because they are the foundation of much else:

$$\mathbb{E}[X] = p, \qquad \text{Var}(X) = p(1 - p).$$

The variance result follows from $\mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p)$, using the fact that $X^2 = X$ for a 0/1 variable. The Bernoulli variance has a feature with real consequences: as a function of p, the quantity $p(1 - p)$ is maximized at $p = 1/2$ and falls to zero at $p = 0$ and $p = 1$. Uncertainty is greatest when the outcome is a coin flip and vanishes as the outcome becomes certain in either direction. Any quantity that is fundamentally a binary outcome — a contract that settles at one of two values, an event that does or does not happen — has $p(1 - p)$ as its variance, and the dependence of that variance on p is the correct way to think about how much is at stake as the probability moves between certainty and a toss-up.

5.1.2 The binomial distribution: counting successes

Repeat a Bernoulli trial independently and count the successes.

Definition 5.2 (Binomial)**.** A **binomial** random variable $X \sim \text{Binomial}(n, p)$ is the number of successes in n independent $\text{Bernoulli}(p)$ trials. Its PMF is

$$p(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

The formula is transparent in light of Chapter 3: any particular sequence with k successes and $n - k$ failures has probability $p^k(1 - p)^{n-k}$, and the binomial coefficient $\binom{n}{k}$ counts how

many such sequences there are. Because a binomial variable is a sum of n independent Bernoullis, its mean and variance follow at once from the additivity results of Chapter 4 — expectations add, and variances add under independence:

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

There is no need to derive these from the PMF; they are simply n times the Bernoulli mean and variance. The binomial governs any setting where one counts the number of times something happens in a fixed number of independent attempts.

5.1.3 The geometric distribution: waiting for success

Instead of fixing the number of trials and counting successes, fix the target at one success and count the trials needed to reach it.

Definition 5.3 (Geometric)**.** A **geometric** random variable $X \sim \mathrm{Geometric}(p)$ is the number of independent $\mathrm{Bernoulli}(p)$ trials up to and including the first success, with PMF $p(k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$

Its mean is $\mathbb{E}[X] = 1/p$, which matches intuition: if successes occur one time in five, you wait five trials on average. Its variance is $(1-p)/p^2$. The geometric distribution is the discrete model for waiting times — how long until the next occurrence — and it possesses the memoryless property discussed below in its continuous cousin, the exponential.

5.1.4 The Poisson distribution: rare events and counts

The Poisson distribution models the number of events occurring in a fixed interval of time or space when events happen independently at a constant average rate.

Definition 5.4 (Poisson)**.** A **Poisson** random variable $X \sim \mathrm{Poisson}(\lambda)$ has PMF

$$p(k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \ldots,$$

where $\lambda > 0$ is the average number of events.

The Poisson has the striking property that its mean and variance are *equal*:

$$\mathbb{E}[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

This equality is itself a useful diagnostic: count data whose variance markedly exceeds its mean is said to be *overdispersed* and is poorly described by a Poisson model, a signal that events are clustering rather than arriving independently. The Poisson arises as a limit of the binomial when n is large and p is small with $np = \lambda$ held fixed — that is, many trials each with a tiny success probability — which is why it is the natural model for rare events: the number of failures in a large system, the number of rare incidents in a period, the number of arrivals in a queue.

5.2 Continuous distributions

5.2.1 The uniform distribution

The **continuous uniform** distribution $X \sim \mathrm{Uniform}(a, b)$ spreads probability evenly over an interval, with constant density $f(x) = 1/(b-a)$ for $x \in [a, b]$ and zero outside. Its mean is the midpoint $(a+b)/2$ and its variance is $(b-a)^2/12$. It is the model of complete ignorance within known bounds, and it plays a special technical role: applying the inverse CDF of any distribution to a $\mathrm{Uniform}(0, 1)$ variable produces a draw from that distribution, which is the basis of how random samples from arbitrary distributions are simulated.

5.2.2 The exponential distribution and memorylessness

The **exponential** distribution $X \sim \text{Exponential}(\lambda)$ is the continuous model for the waiting time until an event that occurs at constant rate λ. Its density is $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, its CDF is $F(x) = 1 - e^{-\lambda x}$, and its mean and variance are

$$\mathbb{E}[X] = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Its defining feature is *memorylessness*: the probability of waiting at least s more units, given that you have already waited t, does not depend on t. The process does not age; having waited a long time tells you nothing about how much longer you must wait. The exponential is the continuous-time companion of the geometric distribution and is the waiting-time distribution associated with Poisson arrivals.

5.2.3 The normal distribution

The normal (or Gaussian) distribution is the most important distribution in statistics, and the reason will become clear in Chapter 6: sums and averages of many independent influences are approximately normal regardless of the distribution of the individual influences. This is why the normal appears wherever a quantity results from the accumulation of many small, independent effects.

Definition 5.5 (Normal)**.** A **normal** random variable $X \sim N(\mu, \sigma^2)$ has the bell-shaped density

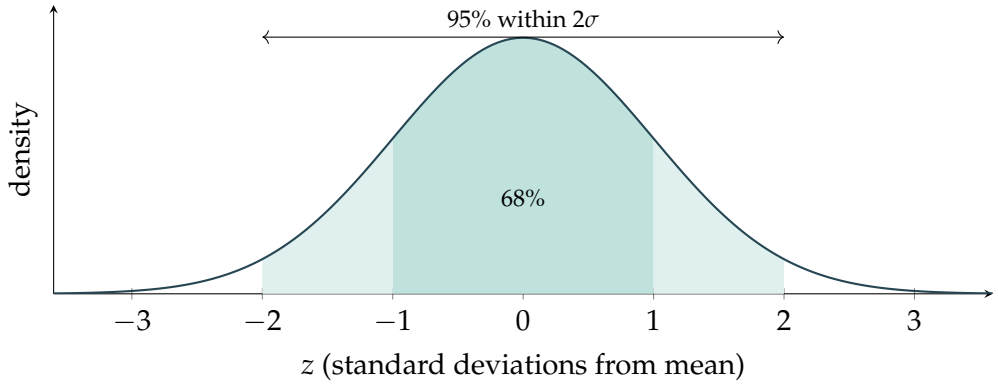
$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R},$$

with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$.

The normal is symmetric about its mean μ, which is therefore also its median and mode, and its spread is governed entirely by σ. Its two parameters are exactly its mean and variance, so specifying those specifies the whole distribution. The **standard normal** is the special case $Z \sim N(0, 1)$, and any normal variable is converted to standard form by the z-score transformation of Chapter 2, $Z = (X - \mu)/\sigma$. This is why a single table or function for the standard normal suffices for all normal calculations: probabilities for any $N(\mu, \sigma^2)$ are obtained by standardizing and reading the standard normal CDF, conventionally written $\Phi(z)$.

The symmetry and known shape give the **68–95–99.7 rule**: about 68% of the probability lies within one standard deviation of the mean, about 95% within two, and about 99.7% within three. This rule is the source of the intuition that a deviation of two standard deviations is "unusual" (a one-in-twenty event in the tails) and three is "rare."

The standard normal and the 68–95–99.7 rule



A further property makes the normal analytically central: a linear combination of independent normal variables is itself normal. If $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ independently, then $aX + bY$ is normal with mean $a\mu_X + b\mu_Y$ and variance $a^2\sigma_X^2 + b^2\sigma_Y^2$. The normal family is closed under the linear operations we perform constantly, which is part of why it is so tractable.

A word of caution, foreshadowed in Chapter 2 and sharpened in Chapter 12, belongs here. The normal distribution has very thin tails — the density decays like $e^{-x^2/2}$, extraordinarily fast. Events beyond a few standard deviations are, under a normal model, essentially impossible. Many real quantities, financial returns prominent among them, have *heavier* tails than this: extreme moves occur far more often than the normal allows. Assuming normality for such a quantity is convenient and often a reasonable first approximation, but it systematically understates the probability of large deviations, and that understatement is precisely where the largest risks hide. The normal is the right default and the wrong place to stop thinking.

5.2.4 The lognormal distribution

Many positive quantities are better modeled not as a sum of small effects but as a *product* of them, and taking logarithms turns products into sums. This motivates the lognormal distribution.

Definition 5.6 (Lognormal)**.** A positive random variable X is **lognormal** if its logarithm is normal: $\ln X \sim N(\mu, \sigma^2)$. Equivalently, $X = e^Y$ where $Y \sim N(\mu, \sigma^2)$.

The lognormal is the natural model whenever a quantity arises from many multiplicative shocks, because by the same logic that makes sums of small effects normal, products of small effects are lognormal. Its mean and variance are *not* simply μ and σ^2 — those are the parameters of the underlying normal, on the log scale. On the original scale,

$$\mathbb{E}[X] = e^{\mu+\sigma^2/2}, \qquad \mathrm{Var}(X) = \left(e^{\sigma^2} - 1\right) e^{2\mu+\sigma^2}.$$

The appearance of the $\sigma^2/2$ term in the mean is a direct instance of the warning from Chapter 4 that $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$ for nonlinear g: the mean of the exponential is not the exponential of the mean. The lognormal is strictly positive (it cannot go below zero, appropriate for prices and other inherently positive quantities) and *right-skewed* with a heavy right tail, the skewness growing with σ. It is the standard model for asset prices in continuous-time finance, precisely because prices are positive and evolve through compounding multiplicative returns; the connection between the lognormal price and the normal log-return is one of the most-used facts in quantitative finance, and it is why log-returns, rather than raw returns, are the standard object of statistical analysis.

5.2.5 The gamma distribution

The **gamma** distribution generalizes the exponential: the sum of k independent Exponential(λ) waiting times is Gamma-distributed, modeling the total time until the k-th event. With shape k and scale $\theta = 1/\lambda$, its mean is $k\theta$ and variance $k\theta^2$. The gamma is a flexible family for positive, right-skewed quantities, and it contains both the exponential (as $k = 1$) and, as we see next, the chi-squared distribution as special cases.

5.3 The sampling distributions

Three further continuous distributions are distinguished by their *role*: they are not primarily models for raw data but the distributions that arise from *functions of normal samples*, and they

are the distributions in which the inferential procedures of Part III are expressed. They are introduced here so that the catalogue is complete; their use will become concrete in Chapters 7 and 8.

5.3.1 The chi-squared distribution

Definition 5.7 (Chi-squared)**.** If $Z_1, \dots, Z_k$ are independent standard normal variables, then $\sum_{i=1}^{k} Z_i^2$ has the **chi-squared** distribution with k *degrees of freedom*, written χ^2_k.

The chi-squared distribution is, by construction, a sum of squared standard normals; it is a special case of the gamma family. It takes only non-negative values and is right-skewed, with mean equal to its degrees of freedom and variance twice that:

$$\mathbb{E}[\chi^2_k] = k, \qquad \mathrm{Var}(\chi^2_k) = 2k.$$

It is the distribution of *sums of squares*, and since variance is built from sums of squared deviations, the chi-squared is the distribution governing the sample variance: for a normal sample, the quantity $(n-1)s^2/\sigma^2$ follows a χ^2_{n-1} distribution. The degrees of freedom being $n-1$ rather than n is the same one-lost-to-estimating-the-mean phenomenon met in Chapter 2. The chi-squared also governs the goodness-of-fit and independence tests of Chapter 8, where observed counts are compared to expected counts through a sum of squared discrepancies.

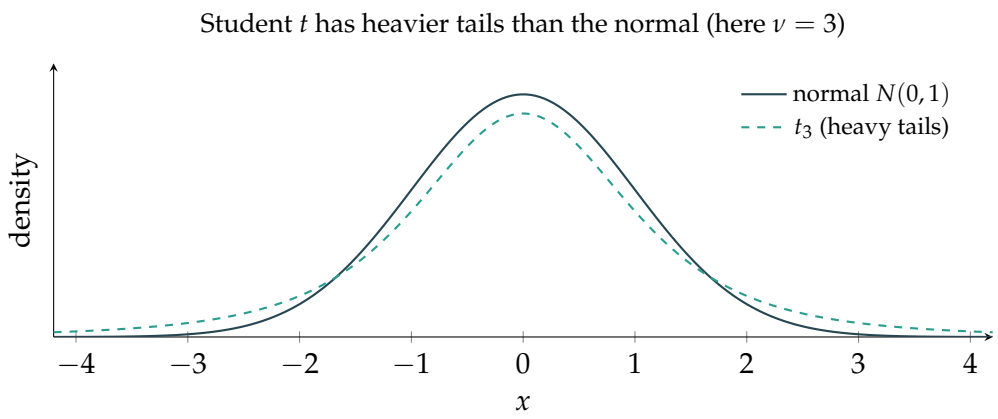
5.3.2 The Student t distribution

Definition 5.8 (Student t)**.** If $Z \sim N(0,1)$ and $V \sim \chi^2_\nu$ are independent, then

$$T = \frac{Z}{\sqrt{V/\nu}}$$

has the **Student** t distribution with ν degrees of freedom.

The t distribution looks like a standard normal — symmetric and bell-shaped, centered at zero — but with *heavier tails*, and the heaviness is controlled by ν. For $\nu > 2$ its variance is $\nu/(\nu-2)$, which exceeds one and shrinks toward one as ν grows; as $\nu \to \infty$ the t distribution converges to the standard normal. The reason the t distribution exists, and the reason it has heavier tails than the normal, is the central practical problem of inference: when we standardize a sample mean we must divide by an *estimated* standard deviation s rather than the true σ, and that estimate is itself uncertain. The extra uncertainty from estimating the spread fattens the tails, and the t distribution captures exactly this. It is the correct distribution for inference about a mean when the variance is unknown and estimated from the data — which is to say, essentially always — and it is therefore the distribution behind the confidence intervals and t-tests of Chapters 7 and 8 and the coefficient tests of regression in Chapters 10 and 11. With small samples the difference from the normal is substantial; with large samples it is negligible, which is why the normal and t procedures coincide as n grows.



5.3.3 The F distribution

Definition 5.9 (F)**.** If $U \sim \chi^2_{d_1}$ and $V \sim \chi^2_{d_2}$ are independent, then

$$F = \frac{U/d_1}{V/d_2}$$

has the F **distribution** with d_1 and d_2 degrees of freedom.

The F distribution is a ratio of two scaled chi-squared variables, hence a ratio of two independent estimates of variance. It is non-negative and right-skewed. Its role is to compare variances and, more importantly for this book, to test whether *several* regression coefficients are jointly zero by comparing the variance explained by a model to the variance left unexplained. The overall significance test of a regression, and every nested-model comparison in Chapter 11, is an F test. The three sampling distributions are tightly linked: a t variable squared is an F with one numerator degree of freedom, and an F with a large denominator degree of freedom approaches a scaled chi-squared. They are three views of the same underlying normal-sampling machinery.

5.4 A map of the distributions

The distributions of this chapter form a connected family rather than a list. The Bernoulli is the atom; summing independent Bernoullis gives the binomial; letting the number of trials grow while the success probability shrinks gives the Poisson. On the continuous side, the exponential models a single waiting time, summing exponentials gives the gamma, and the chi-squared is the gamma that arises from squared normals. The normal sits at the center by virtue of the Central Limit Theorem, with the lognormal as its multiplicative image under the logarithm. And the three sampling distributions are all manufactured from normals: the chi-squared from sums of squares, the t from a normal divided by a square-rooted chi-squared, and the F from a ratio of chi-squareds. Recognizing which mechanism is at work in a given problem tells you which distribution applies, and the relationships let you move between them. The next chapter explains why the normal occupies the center of this map: the Central Limit Theorem, the result that makes the normal the universal distribution of averages and the foundation of all that follows.

Part III

Inference

Chapter 6

Sampling Distributions and the Central Limit Theorem

We now arrive at the conceptual hinge of the entire subject. Parts I and II gave us tools to describe data and to model randomness. This chapter explains why those tools allow inference at all — why a single finite sample can support quantified statements about an unknown population. The answer rests on one idea, the *sampling distribution*, and one theorem, the Central Limit Theorem, which together are the reason statistics works.

6.1 A statistic is a random variable

Here is the shift in perspective that unlocks inference. A statistic such as the sample mean $\bar{X}$ is computed from data, and the data are the result of random sampling. Therefore the statistic is *itself random*: a different sample would have produced a different value. This is obvious once stated, but its consequences are profound. If we drew many independent samples of size n from the same population and computed $\bar{X}$ for each, we would obtain a whole collection of values that vary from sample to sample. That collection has a distribution.

Definition 6.1 (Sampling distribution)**.** The **sampling distribution** of a statistic is the probability distribution of the values the statistic takes across all possible samples of a given size n drawn from the population.

The sampling distribution is the central object of inference, and it is essential to keep it distinct from two other distributions it is easily confused with. It is not the distribution of the *population* (the distribution of individual values), and it is not the distribution of the *sample* (the histogram of the one dataset in hand). It is the distribution of a *statistic* computed from samples — a distribution one rarely observes directly but can characterize mathematically. Once we know the sampling distribution of a statistic, we know how much that statistic varies from sample to sample, and that is precisely the quantity needed to attach error bars to an estimate or to judge whether an observed value is surprising. The entire machinery of Parts III and IV consists of working out, or approximating, the sampling distributions of various statistics.

6.2 The sampling distribution of the mean

The most important statistic is the sample mean, and its sampling distribution has two features we can pin down exactly, both derived in Chapter 4 from the algebra of sums. Suppose the population has mean μ and variance σ^2, and we draw an independent sample $X_1, \dots, X_n$. The sample mean $\bar{X} = \frac{1}{n}\sum_i X_i$ then has

$$\mathbb{E}[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

These three facts deserve to be internalized. The first says the sampling distribution of the mean is *centered on the truth*: averaged over all possible samples, the sample mean neither over- nor under-shoots the population mean. A statistic with this property is called *unbiased*, and unbiasedness is the formal expression of the idea that the estimator is correct on average. The second and third say that the *spread* of the sampling distribution shrinks as the sample grows — and shrinks specifically in proportion to $1/\sqrt{n}$.

Definition 6.2 (Standard error)**.** The **standard error** of a statistic is the standard deviation of its sampling distribution. For the sample mean, the standard error is $\sigma/\sqrt{n}$.

The distinction between the *standard deviation* and the *standard error* is one of the most important and most frequently muddled in all of applied statistics, so it is worth stating slowly. The standard deviation σ measures the spread of *individual observations* in the population — how much one data point typically differs from the mean. The standard error $\sigma/\sqrt{n}$ measures the spread of the *sample mean* — how much an *average* of n observations typically differs from the population mean. The standard error is smaller, by a factor of $\sqrt{n}$, because averaging cancels out idiosyncratic fluctuations. When you report "the average is X plus or minus something," the "something" is a standard error, not a standard deviation; confusing the two either drastically overstates or drastically understates your uncertainty.

The $1/\sqrt{n}$ scaling has a sobering practical implication: precision improves only with the *square root* of sample size. To halve the standard error you must *quadruple* the data; to cut it to a tenth you need a hundred times as much. This diminishing return on data is a fundamental and inescapable feature of estimation, and it sets a hard ceiling on how much certainty any finite sample can buy.

6.3 The Law of Large Numbers

The shrinking standard error already suggests a limiting result: as n grows without bound, the sampling distribution of the mean concentrates ever more tightly on μ. This is the Law of Large Numbers.

Theorem 6.1 (Weak Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be independent and identically distributed with mean μ and finite variance. Then the sample mean converges in probability to μ: for any $\varepsilon > 0$,*

$$\mathbb{P}\left(|\bar{X}_n - \mu| > \varepsilon \right) \to 0 \quad \text{as } n \to \infty.$$

The Law of Large Numbers is the formal guarantee that averaging works: with enough data, the sample mean is arbitrarily likely to be arbitrarily close to the true mean. It is the theorem that justifies estimating a population mean by a sample mean in the first place, and it underwrites the frequency interpretation of probability — that the long-run relative frequency of an event converges to its probability, since a relative frequency is just the sample mean of a $0/1$ indicator. But the Law of Large Numbers, for all its importance, tells us only *where* the sample mean ends up. It says nothing about the *shape* of its distribution along the way, and it is the shape that we need for inference. That is supplied by a deeper and more surprising result.

6.4 The Central Limit Theorem

The Central Limit Theorem is the most important theorem in statistics, and one of the most remarkable results in all of mathematics. It states that the sampling distribution of the mean is approximately *normal*, almost regardless of the shape of the population from which the data are drawn.

Theorem 6.2 (Central Limit Theorem)**.** *Let $X_1, \ldots, X_n$ be independent and identically distributed with mean μ and finite variance σ^2. Then as $n \to \infty$, the standardized sample mean converges in distribution to a standard normal:*

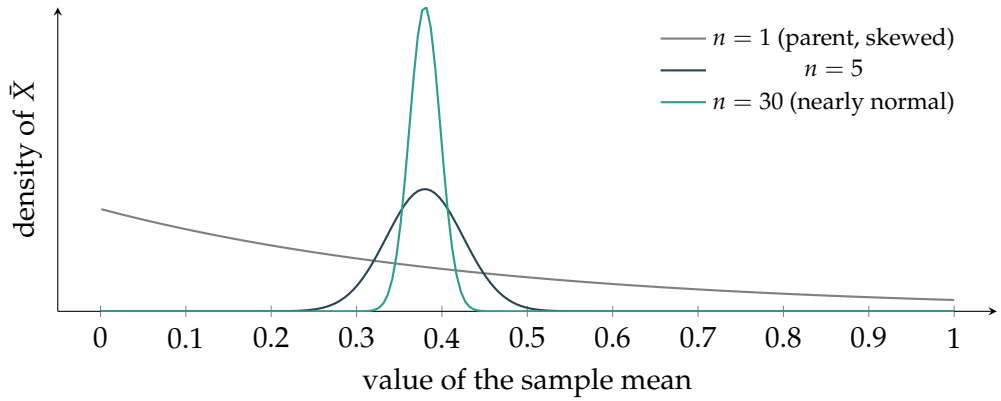
$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

Equivalently, for large n the sample mean is approximately $N\left(\mu,\ \sigma^2/n\right)$.

Pause over what this claims. The individual observations may follow almost *any* distribution — skewed, bimodal, discrete, heavy in the middle, bounded, whatever — and yet the distribution of their *average* is approximately normal, with the approximation improving as n grows. The population's own shape is washed away by the act of averaging; only its mean and variance survive, as the two parameters of the limiting normal. This is why the normal distribution is ubiquitous in nature and in data: any quantity that is itself the sum or average of many small, independent contributions will be approximately normal regardless of the nature of those contributions. Measurement errors, aggregate quantities, and averages of all kinds inherit normality from the Central Limit Theorem rather than from any normality in their parts.

The practical importance is just as great. The Central Limit Theorem is what makes inference about a mean possible *without knowing the population distribution*. We almost never know the shape of the population, but the theorem tells us that we do not need to: the sampling distribution of the mean is approximately normal regardless, and that approximate normality is exactly what the confidence intervals of Chapter 7 and the tests of Chapter 8 are built on. Without the Central Limit Theorem, every inference about a mean would require knowing the population's full distribution; with it, knowing the mean and variance suffices.

Averaging a skewed parent: the sampling distribution of $\bar{X}$ becomes normal as n grows



6.4.1 What the theorem does and does not say

Three qualifications keep the Central Limit Theorem from being misapplied. First, it concerns the distribution of the *mean* (or sum), not the distribution of the individual data. Averaging does not make your raw data normal; a histogram of skewed observations stays skewed no matter how many you collect. The theorem is a statement about $\bar{X}$, not about the X_i. Second, it requires *finite variance*. Distributions with tails so heavy that their variance is infinite do not obey the standard Central Limit Theorem, and averages of such variables do not become normal — a caveat with teeth in finance, where some quantities are heavy-tailed enough that this assumption is genuinely in doubt. Third, it is an *asymptotic* result, exact only in the limit; for finite n it is an approximation whose quality depends on the situation.

6.4.2 How large must n be?

The quality of the normal approximation for finite n depends on how non-normal the population is. A common rule of thumb holds that $n \geq 30$ suffices, and for populations that are already roughly symmetric and light-tailed, the approximation is excellent at much smaller n. But the rule is only a rule of thumb, and it can fail badly. The more skewed or heavy-tailed the population, the larger the sample needed before the sampling distribution of the mean looks normal; for severely skewed or heavy-tailed data, even several hundred observations may be insufficient, and the residual skewness in the sampling distribution can distort inferences in ways that matter. The honest practice is to treat $n \geq 30$ as a starting presumption, not a guarantee, and to be especially wary when the underlying data are visibly skewed or prone to extreme values — exactly the conditions, again, that prevail in much financial data.

6.5 The Central Limit Theorem for proportions

A proportion is a special case of a mean, and so the Central Limit Theorem applies to it directly — a fact used throughout polling, quality control, and any setting involving rates. If each observation is a Bernoulli(p) indicator, the sample proportion $\hat{p} = \frac{1}{n}\sum_i X_i$ is literally the sample mean of $0/1$ variables. Inheriting the general results with the Bernoulli mean p and variance $p(1-p)$, we get

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}, \qquad \hat{p} \approx N\left(p,\ \frac{p(1-p)}{n}\right)$$

for large n. The standard error of a proportion is thus $\sqrt{p(1-p)/n}$. The approximation is good when n is large enough that both np and $n(1-p)$ are reasonably sized (a common threshold is that both exceed about 10); it degrades when p is very close to 0 or 1, because the underlying Bernoulli is then so asymmetric that the sampling distribution of $\hat{p}$ remains skewed unless n is very large. This is the same caution as before, specialized to the binary case, and it is the foundation of the confidence intervals and tests for proportions in the next two chapters.

6.6 Sampling distributions of other statistics

The mean is not the only statistic with a known sampling distribution, and the others connect directly to the sampling distributions of Chapter 5. When the population is normal, the sample variance has a sampling distribution too: the scaled quantity $(n-1)s^2/\sigma^2$ follows a chi-squared distribution with $n-1$ degrees of freedom. This is the basis for inference about a variance, and the degrees of freedom are again $n-1$ because one was spent estimating the mean. More importantly for practice, when we standardize the sample mean but must use the *estimated* standard deviation s in place of the unknown σ, the resulting quantity

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

does not follow a standard normal but a Student t distribution with $n-1$ degrees of freedom, for exactly the reason given in Chapter 5: the extra uncertainty from estimating σ fattens the tails. This is the single most important sampling-distribution fact for applied work, because σ is essentially never known, and it is why the t distribution, not the normal, is the workhorse of inference about means in Chapters 7, 8, 10, and 11. For large n the t and the normal coincide, consistent with the Central Limit Theorem, so the two approaches agree when data are plentiful and diverge — with the t being correct — when data are scarce.

6.7 Summary

A statistic is a random variable, and its sampling distribution — the distribution of its values across repeated samples — is the object that makes inference possible. For the sample mean, that distribution is centered on the true mean (unbiasedness) with spread given by the standard error $\sigma/\sqrt{n}$, a quantity that must be sharply distinguished from the population standard deviation and that improves only as the square root of sample size. The Law of Large Numbers guarantees the sample mean converges to the truth; the Central Limit Theorem, far more powerfully, guarantees that its distribution becomes normal regardless of the population's shape, provided the variance is finite. This is why the normal distribution sits at the center of the map and why inference about a mean needs only the mean and variance, not the full population distribution. When σ must be estimated, the relevant distribution becomes the Student t, the correction for that added uncertainty. With sampling distributions understood, we can finally do what the whole subject was building toward: estimate unknown parameters and quantify how wrong we might be. That is the work of Chapter 7.

Chapter 7

Estimation and Confidence Intervals

The Central Limit Theorem told us how a sample mean is distributed around the population mean. We now put that knowledge to work on the central task of inference: using a sample to produce a numerical guess for an unknown parameter, and — just as importantly — an honest statement of how uncertain that guess is. The first goal is *point estimation*, producing a single best number; the second is *interval estimation*, producing a range that plausibly contains the truth. The two together are how statistics answers a quantitative question without pretending to a certainty it does not have.

7.1 Point estimation

Definition 7.1 (Estimator and estimate)**.** An **estimator** is a rule or formula, a function of the sample, used to approximate an unknown parameter θ; we write it $\hat{\theta}$. The **estimate** is the particular numerical value the estimator takes on a given sample.

The distinction matters: the estimator $\hat{\theta}$ is a random variable with a sampling distribution (Chapter 6), while the estimate is one realization of it. The sample mean $\bar{X}$ is an estimator of the population mean μ; the number 9.4 it produces on a particular dataset is an estimate. To judge whether an estimator is any good, we ask how its sampling distribution sits relative to the truth, and three properties capture the answer.

7.1.1 Bias, variance, and mean squared error

Definition 7.2 (Bias)**.** The **bias** of an estimator is the difference between its expected value and the true parameter,

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

An estimator is **unbiased** if its bias is zero, $\mathbb{E}[\hat{\theta}] = \theta$.

Unbiasedness says that the estimator is correct *on average* across repeated samples — it does not systematically over- or under-shoot. The sample mean is unbiased for the population mean, and the sample variance s^2 with its $n-1$ divisor is, by construction, unbiased for σ^2; that divisor (Chapter 2) is exactly what removes the bias that dividing by n would introduce. But unbiasedness alone is not enough. An estimator can be unbiased and yet so variable that any single estimate is far from the truth. We must also consider the estimator's *variance*, $\text{Var}(\hat{\theta})$, which measures how much its value bounces around from sample to sample.

The two concerns — being right on average and being stable — are combined in a single criterion.

Definition 7.3 (Mean squared error)**.** The **mean squared error (MSE)** of an estimator is the expected squared distance from the truth,

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \theta)^2\big] = \text{Var}(\hat{\theta}) + \big[\text{Bias}(\hat{\theta})\big]^2.$$

The decomposition on the right — that mean squared error equals variance plus squared bias — is one of the most important identities in all of statistics, and it follows from adding and subtracting $\mathbb{E}[\hat{\theta}]$ inside the square and expanding. It says the total error of an estimator has exactly two sources: how far off it is on average (bias) and how much it varies (variance). This decomposition is the formal statement of the **bias–variance tradeoff**, a theme that runs through the whole of statistics and machine learning. It is often possible to reduce variance at the cost of introducing a little bias, and if the variance reduction is large enough, the biased estimator has lower total error and is genuinely better. Insisting on unbiasedness for its own sake can therefore be a mistake; what we ultimately care about is small mean squared error, and a slightly biased but much more stable estimator can dominate an unbiased but volatile one. This single idea justifies a vast amount of modern practice, from regularized regression to shrinkage estimation.

7.1.2 Consistency and efficiency

Two further properties describe how estimators behave as data accumulate and relative to one another. An estimator is **consistent** if it converges to the true parameter as the sample size grows without bound — that is, its sampling distribution concentrates on θ as $n \to \infty$. The Law of Large Numbers (Chapter 6) is precisely the statement that the sample mean is a consistent estimator of the population mean. Consistency is a minimal requirement: an estimator that does not approach the truth even with unlimited data is hard to defend. Among consistent, unbiased estimators, we prefer the one with the smallest variance; such an estimator is called **efficient**. Efficiency formalizes the idea of extracting the most information from a given sample, and there is a theoretical floor on the variance any unbiased estimator can achieve, so an efficient estimator is one that attains the best possible precision.

7.2 How to construct estimators

Two general-purpose recipes produce good estimators in a principled way, rather than by guesswork.

7.2.1 The method of moments

The **method of moments** is the older and simpler recipe: set the moments of the population equal to the corresponding moments of the sample, and solve for the parameters. To estimate the mean, set the population mean equal to the sample mean; to estimate a two-parameter distribution, match the first two moments (mean and variance) and solve the resulting equations. The method is intuitive and easy to apply, and it often gives a reasonable starting estimate, though it is not always efficient and can occasionally produce estimates outside the sensible range of a parameter.

7.2.2 Maximum likelihood estimation

The dominant method in modern statistics is **maximum likelihood**, which rests on a compelling principle: choose the parameter value that makes the observed data most probable. Given a model with parameter θ and independent observations $x_1, \dots, x_n$, the **likelihood** is the probability (or density) of the observed data viewed as a function of θ,

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta),$$

a product because the observations are independent. The **maximum likelihood estimator (MLE)** is the value $\hat{\theta}$ that maximizes $L(\theta)$. Because products are awkward to differentiate and prone to numerical underflow, one almost always maximizes the **log-likelihood** instead,

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta),$$

which has the same maximizer because the logarithm is increasing but turns the product into a far more tractable sum. The maximum is then found by setting the derivative of ℓ to zero and solving.

Maximum likelihood is the workhorse of statistical estimation for good reason: under broad conditions the MLE has excellent large-sample properties. It is consistent, it is *asymptotically efficient* (achieving the theoretical minimum variance in large samples), and it is *asymptotically normal*, meaning that for large n the estimator is approximately normally distributed around the true value — which is exactly what makes it possible to build confidence intervals and tests around maximum likelihood estimates. As a concrete and reassuring example, when the model is normal, the maximum likelihood estimator of the mean turns out to be exactly the sample mean, so the principled method recovers the obvious estimator; and the least-squares estimators at the center of Chapters 10 and 11 coincide with maximum likelihood estimators under the assumption of normal errors, which is part of why least squares is so well motivated.

7.3 Interval estimation: confidence intervals

A point estimate alone is incomplete, even dishonest, because it conveys no sense of its own reliability. The estimate 9.4 means something very different when it is pinned down to within ± 0.1 than when it could plausibly be anywhere from 5 to 14. A *confidence interval* attaches that missing measure of precision by reporting a range of plausible values for the parameter.

Definition 7.4 (Confidence interval)**.** A **confidence interval** with confidence level $1 - \alpha$ is an interval, computed from the sample, constructed by a procedure that contains the true parameter in a proportion $1 - \alpha$ of all possible samples.

The interpretation of a confidence interval is notoriously subtle and is worth stating with care, because the natural reading is wrong. A 95% confidence interval does *not* mean there is a 95% probability that the true parameter lies in this particular interval. The true parameter is a fixed (if unknown) number; it is either in the interval or it is not, and there is no probability about it once the interval is computed. What the 95% refers to is the *procedure*: if we repeated the sampling and interval construction many times, about 95% of the resulting intervals would contain the true parameter. The confidence is in the long-run reliability of the method, not in any single interval. The randomness lives in the interval's endpoints, which vary from sample to sample, not in the fixed parameter. Keeping this straight is the difference between understanding confidence intervals and merely reciting them.

7.3.1 Confidence interval for a mean, variance known

The construction follows directly from the sampling distribution of the mean. By the Central Limit Theorem, $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$, so the standardized mean $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ is approximately standard normal. Since a standard normal falls within $\pm z_{\alpha/2}$ with probability $1 - \alpha$, where $z_{\alpha/2}$ is the standard normal critical value (equal to 1.96 for the common 95% level), a little algebra rearranges this statement into an interval for μ:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

The quantity added and subtracted, $z_{\alpha/2}\,\sigma/\sqrt{n}$, is the **margin of error**. Its structure is illuminating: it is a critical value (how many standard errors, set by the desired confidence) times the standard error (the spread of the sampling distribution). The interval is wider when we demand more confidence (larger $z_{\alpha/2}$), wider when the data are more variable (larger σ), and narrower when we have more data (larger n, through the $\sqrt{n}$ in the denominator). All the intuition about precision from Chapter 6 is visible in this one formula.

7.3.2 Confidence interval for a mean, variance unknown

In practice σ is essentially never known and must be estimated by the sample standard deviation s. As established in Chapters 5 and 6, replacing σ by s changes the relevant distribution from the normal to the Student t with $n-1$ degrees of freedom, because the added uncertainty of estimating the spread fattens the tails. The interval becomes

$$\bar{X} \pm t_{\alpha/2,\,n-1}\,\frac{s}{\sqrt{n}},$$

where $t_{\alpha/2,n-1}$ is the critical value from the t distribution with $n-1$ degrees of freedom. Because the t distribution has heavier tails, its critical values are larger than the corresponding normal ones, making the interval appropriately wider to reflect our ignorance of σ. For small samples this widening is substantial and essential; for large samples the t critical value approaches the normal one and the two intervals nearly coincide. This t-interval is the standard confidence interval for a mean in real applications, and it is the direct ancestor of the coefficient intervals in regression.

7.3.3 Confidence intervals for proportions and variances

The same logic produces intervals for other parameters. For a **proportion**, using the normal approximation to the sampling distribution of $\hat{p}$ from Chapter 6 with standard error $\sqrt{\hat{p}(1-\hat{p})/n}$, the interval is

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}},$$

valid when n is large enough for the normal approximation to hold. For a **variance**, the interval is built not from a symmetric $\pm$ form but from the chi-squared distribution, because $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ and the chi-squared is asymmetric; the resulting interval for σ^2 has the sample variance scaled by ratios involving the upper and lower chi-squared critical values, and it is correspondingly lopsided. The unifying idea across all these cases is identical: identify the sampling distribution of the relevant statistic, find the values that bracket the central $1-\alpha$ of that distribution, and invert to obtain a range for the parameter.

7.3.4 Determining sample size

The margin-of-error formula can be run in reverse to answer a question of real practical value: how much data is needed to achieve a desired precision? Setting the margin of error for a mean equal to a target value E and solving for n gives

$$n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2.$$

The squared dependence is the now-familiar $1/\sqrt{n}$ law seen from the other side: halving the target margin of error quadruples the required sample size. This calculation is how studies and data-collection efforts are planned, and it makes the cost of precision explicit before any data are gathered.

7.4 Summary

Estimation produces both a best guess and an honest account of its uncertainty. A point estimator is judged by its bias, its variance, and the mean squared error that combines them through the fundamental decomposition MSE $=$ variance $+$ bias^2, which formalizes the pervasive bias–variance tradeoff; consistency and efficiency describe its limiting and relative behavior. Estimators are constructed principledly by the method of moments or, more powerfully, by maximum likelihood, whose asymptotic normality is what makes inference around it possible. Confidence intervals convert the sampling distribution of an estimator into a range of plausible parameter values, with the critical caveat that the confidence level describes the long-run reliability of the procedure, not the probability for any single interval. The mean interval takes the z form when σ is known and the wider t form when it is estimated, and the same construction, applied through the appropriate sampling distribution, yields intervals for proportions and variances. The next chapter turns the same machinery to a different question: not "what is the parameter?" but "is the parameter consistent with a specific claim?" — the logic of hypothesis testing.

Chapter 8

Hypothesis Testing

Confidence intervals answer "what is the parameter, and how precisely do we know it?" A closely related but distinct question is "is the parameter consistent with a specific claim?" This is the domain of hypothesis testing, the formal procedure for deciding whether the data provide enough evidence to reject a stated position. It is among the most used — and most misused — tools in all of quantitative work, so this chapter develops both the mechanics and the interpretation with care.

8.1 The logic of testing

Hypothesis testing adopts a stance borrowed from the logic of falsification. We do not set out to prove a claim true; we set out to see whether the data are inconsistent enough with a default position to warrant abandoning it. This asymmetric structure is encoded in two competing hypotheses.

Definition 8.1 (Null and alternative hypotheses)**.** The **null hypothesis** H_0 is the default or status-quo claim, typically a statement of "no effect" or "no difference," which is presumed true unless the evidence is strong enough to overturn it. The **alternative hypothesis** H_1 (or H_a) is the claim we would conclude if the null is rejected.

The null hypothesis plays the role of the presumption of innocence: it stands until the evidence against it passes a high bar. We never "accept" or "prove" the null; we either *reject* it (the evidence is strong enough) or *fail to reject* it (the evidence is insufficient). Failing to reject is emphatically not the same as proving the null true — it means only that we did not find enough evidence to abandon it, just as an acquittal is not a proof of innocence. This careful language is not pedantry; it guards against the pervasive error of treating a non-significant result as positive evidence for the null.

The alternative may be *two-sided*, asserting merely that the parameter differs from the null value in either direction (for instance $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$), or *one-sided*, asserting a difference in a specific direction ($H_1 : \mu > \mu_0$). The choice must be made before seeing the data and on substantive grounds; switching to a one-sided test after observing the direction of the effect is a form of cheating that invalidates the stated error rate.

8.2 Test statistics and the two kinds of error

A test works by computing a **test statistic** from the sample — a number measuring how far the data deviate from what the null hypothesis would predict — and then asking how surprising that value would be *if the null were true*. The key is that, under the null hypothesis, the test statistic has a known sampling distribution (drawn from the distributions of Chapters 5 and 6), so we can quantify exactly how extreme the observed value is.

Because the decision is made under uncertainty, two distinct errors are possible, and they are not symmetric.

Definition 8.2 (Type I and Type II errors)**.** A **Type I error** is rejecting the null hypothesis when it is in fact true (a false positive). A **Type II error** is failing to reject the null hypothesis when the alternative is in fact true (a false negative). Their probabilities are denoted

$$\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}), \qquad \beta = \mathbb{P}(\text{fail to reject } H_0 \mid H_1 \text{ true}).$$

The probability α of a Type I error is called the **significance level** of the test, and it is chosen in advance — conventionally 0.05, sometimes 0.01. It is the rate at which we are willing to falsely cry wolf. We control α directly by setting the threshold of evidence required to reject: a test rejects the null when the test statistic falls into the *rejection region*, the set of values so extreme that they would occur with probability only α if the null were true. The quantity $1 - \beta$ is the **power** of the test, the probability of correctly detecting a real effect.

Definition 8.3 (Power)**.** The **power** of a test is $1 - \beta$, the probability of rejecting the null hypothesis when the alternative is true. It measures the test's ability to detect a genuine effect.

The two error rates are in tension. Lowering α — demanding stronger evidence before rejecting — makes Type I errors rarer but, all else equal, raises β and lowers power, because a more conservative test also misses more real effects. The only way to reduce both simultaneously is to gather more data: power increases with sample size, with the size of the true effect, and with lower variability, and it falls as the significance level is made more stringent. Understanding the determinants of power is essential, because a test with low power is nearly worthless — it will usually fail to detect even real effects, so a non-significant result from an underpowered test tells you almost nothing. The most common silent failure in applied testing is running an underpowered test and over-interpreting its failure to reject.

8.3 The p-value

Rather than only reporting a binary reject/fail-to-reject decision, it is standard to report a *p-value*, which conveys the strength of the evidence on a continuous scale.

Definition 8.4 (p-value)**.** The **p-value** is the probability, computed assuming the null hypothesis is true, of obtaining a test statistic at least as extreme as the one actually observed.

A small p-value means the observed data would be very unlikely under the null hypothesis, and therefore constitutes strong evidence against it; the decision rule is to reject the null when the p-value is below the chosen significance level α. The p-value is, in effect, the smallest significance level at which the data would lead to rejection.

The p-value is so routinely misinterpreted that its correct meaning must be stated alongside the errors it invites. The p-value is *not* the probability that the null hypothesis is true — that is a fundamentally different quantity, and the p-value says nothing directly about it. It is the probability of the *data* (or something more extreme) *given the null*, not the probability of the *null given the data*; conflating the two is the same inversion fallacy that Bayes' theorem corrected in Chapter 3. Two further cautions are essential. First, a p-value is not a measure of the *size* or *importance* of an effect: with a large enough sample, a trivially small and practically meaningless effect will produce a tiny p-value, because statistical significance measures only whether an effect is distinguishable from zero, not whether it is large enough to matter. The distinction between **statistical significance** and **practical significance** is one that careful analysts never lose sight of, and it is why effect sizes and confidence intervals should always accompany p-values. Second, the logic of the p-value holds only for a *single pre-specified test*. If many tests are run and only the significant ones reported, or if the hypothesis is chosen after seeing the data, the nominal error rate is meaningless — running twenty independent tests at the 0.05 level produces, on average, one false positive even when nothing is going on. This

multiple comparisons problem is a major source of spurious findings, and guarding against it, by adjusting thresholds or pre-registering hypotheses, is a mark of statistical seriousness.

8.3.1 Tests and confidence intervals are dual

Hypothesis tests and confidence intervals are two views of the same underlying sampling distribution, and the connection is exact and useful. A two-sided test of $H_0 : \mu = \mu_0$ at level α rejects the null if and only if the value μ_0 falls *outside* the $1 - \alpha$ confidence interval for μ. The confidence interval is precisely the set of null values that would *not* be rejected. This duality means a confidence interval contains strictly more information than a test: it not only tells you whether a particular value is rejected but displays the entire range of values consistent with the data, which is one more reason to prefer reporting intervals over bare p-values.

8.4 Tests for means: the z-test and t-test

The most common tests concern means, and they instantiate the general logic with the specific sampling distributions already developed.

The **one-sample test** asks whether a population mean equals a hypothesized value μ_0. The test statistic measures the standardized distance of the sample mean from μ_0. When the population variance is known (a textbook idealization), the statistic

$$z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

follows a standard normal distribution under the null, giving the *z-test*. When the variance is unknown and estimated by s — the realistic case — the statistic

$$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

follows a Student t distribution with $n - 1$ degrees of freedom under the null, giving the *t-test*. The replacement of σ by s and of the normal by the t is the same correction met in Chapter 7, and it is why the t-test, not the z-test, is what one actually runs.

The **two-sample t-test** asks whether two populations have equal means, comparing the difference of sample means $\bar{X}_1 - \bar{X}_2$ to its standard error. This is the standard tool for comparing a treatment group to a control, or one regime to another. A distinct and often more powerful variant is the **paired t-test**, used when the two samples are naturally matched — the same units measured under two conditions, or two series observed at the same times. Pairing works by reducing the comparison to a one-sample test on the *differences* within each pair, which cancels out the variation between units and isolates the effect of interest, typically yielding much greater power than treating the two samples as independent. Recognizing when data are paired, and exploiting the pairing, is a common way to extract a real effect that an unpaired analysis would miss.

8.5 Tests for categorical data: chi-squared tests

When the data are counts in categories rather than measurements, the relevant tests are built on the chi-squared distribution of Chapter 5, and they share a common form: compare *observed* counts O to the *expected* counts E predicted under the null, through a sum of squared, scaled discrepancies,

$$\chi^2 = \sum \frac{(O - E)^2}{E},$$

which follows a chi-squared distribution under the null with degrees of freedom determined by the number of categories. A large value indicates that observed and expected counts diverge more than chance would allow, leading to rejection.

The **goodness-of-fit test** uses this statistic to ask whether a single categorical variable follows a hypothesized distribution — whether observed frequencies across categories match a claimed set of proportions. The **test of independence** applies the same machinery to a two-way contingency table to ask whether two categorical variables are associated: the expected counts are computed under the assumption of independence (using the row and column totals), and a large chi-squared statistic indicates that the variables are in fact related. These tests are the categorical-data counterparts of the mean comparisons above, and they extend the testing framework from measurements to frequencies.

8.6 Comparing variances: the F-test

Finally, to test whether two populations have equal *variances*, the natural statistic is the *ratio* of the two sample variances, which under the null of equal variances follows the F distribution of Chapter 5 with degrees of freedom $n_1 - 1$ and $n_2 - 1$:

$$F = \frac{s_1^2}{s_2^2}.$$

A ratio far from one — in either direction — is evidence that the variances differ. The F-test for variances is somewhat sensitive to non-normality, a limitation worth remembering, but the deeper importance of the F statistic is the role it plays in regression, where exactly this idea of comparing a ratio of variances becomes the test for the joint significance of a model's coefficients in Chapters 10 and 11. The hypothesis-testing logic developed here — a statistic, its null distribution, a rejection region, an error rate — is identical in that setting; only the statistic changes.

8.7 Summary

Hypothesis testing decides whether data are inconsistent enough with a default claim to reject it, using a null hypothesis presumed true until strong evidence overturns it. The procedure trades off two errors — false positives at rate α, the chosen significance level, and false negatives at rate β, with $1 - \beta$ the power — and only more data can reduce both at once, which is why underpowered tests render non-significant results nearly uninformative. The p-value measures the strength of evidence against the null as the probability of data this extreme under the null, and must never be misread as the probability the null is true, as a measure of effect size, or as valid under undisclosed multiple testing. Tests and confidence intervals are dual descriptions of the same sampling distribution. The specific tests — the z- and t-tests for means, paired tests for matched data, chi-squared tests for categorical counts, and the F-test for variances — all instantiate the one logic with the appropriate distribution. That same logic carries directly into the study of relationships between variables, which begins in the next chapter with covariance and correlation and culminates in regression.

Part IV

Relationships and Regression

Chapter 9

Covariance and Correlation

Until now we have studied variables mostly one at a time. The questions with the most practical bite, however, concern how two variables move *together*: when one is high, is the other tend to be high, or low, or neither? Covariance and correlation are the basic measures of this co-movement, and they are the conceptual bridge to regression, the subject of the remaining chapters. They also carry one of the most important warnings in all of applied statistics, about the gap between association and causation.

9.1 Covariance

We met covariance briefly in Chapter 4, in the formula for the variance of a sum. We now study it in its own right.

Definition 9.1 (Covariance)**.** The **covariance** of two random variables X and Y, with means μ_X and μ_Y, is the expected product of their deviations from their means,

$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The second form, obtained by expanding the product and using linearity, is the computational counterpart of the variance identity from Chapter 4, and indeed covariance generalizes variance: $\operatorname{Cov}(X, X) = \operatorname{Var}(X)$. The sign of the covariance carries the essential information. It is *positive* when the two variables tend to lie on the same side of their means together — when X is above average, Y tends to be above average too, so the product of deviations is usually positive. It is *negative* when they tend to lie on opposite sides — high X paired with low Y — so the product of deviations is usually negative. A covariance near zero indicates no consistent linear tendency to co-move. For a *sample*, the covariance is estimated by the natural analogue,

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}),$$

with the same $n - 1$ divisor and for the same degrees-of-freedom reason as the sample variance.

Covariance is bilinear, meaning it distributes over sums and pulls out constants in each argument: $\operatorname{Cov}(aX + b, cY + d) = ac \operatorname{Cov}(X, Y)$. The additive constants b and d vanish — shifting a variable does not change how it co-moves — while the multiplicative constants a and c pass through and multiply the covariance. This last fact exposes the central weakness of covariance as a descriptive measure: *its magnitude depends on the units of the variables*. Measuring a quantity in cents rather than dollars multiplies every covariance involving it by a hundred, without anything about the underlying relationship having changed. The sign of the covariance is meaningful, but its magnitude is not comparable across different pairs of variables or different units, and so a raw covariance of, say, 4,000 tells us a relationship is positive but nothing about how strong it is. To get an interpretable measure of strength, we must remove the units.

9.2 Correlation

Standardizing the covariance by the standard deviations of the two variables removes the units and rescales the measure to a fixed, interpretable range.

Definition 9.2 (Pearson correlation)**.** The **Pearson correlation coefficient** of X and Y is the covariance divided by the product of the standard deviations,

$$\rho_{XY} = \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \, \sigma_Y},$$

and always satisfies $-1 \leq \rho_{XY} \leq 1$.

Correlation is, in effect, the covariance of the standardized variables — the covariance of the z-scores of Chapter 2 — which is why it is unitless and bounded. Its interpretation is clean and is the reason it, rather than covariance, is the everyday measure of association. The *sign* indicates the direction of the linear relationship, exactly as for covariance. The *magnitude* now indicates the strength of the *linear* relationship on a universal scale: a correlation of $+1$ means the points lie exactly on an upward-sloping straight line, -1 means they lie exactly on a downward-sloping line, and 0 means there is no linear relationship at all. Values in between interpolate, with magnitudes near 1 indicating a tight linear association and magnitudes near 0 a weak one. The sample correlation r is computed by the same formula from sample covariance and sample standard deviations, and it is the single most reported number describing the relationship between two quantities. Because it is invariant to separate linear rescalings of either variable, the correlation between two quantities is the same whatever units they are measured in — the property covariance lacked.

9.3 The two great cautions

Correlation is so useful and so heavily used that its two fundamental limitations must be stated as emphatically as its definition. Misunderstanding either is the source of a large fraction of bad quantitative conclusions.

9.3.1 Correlation measures only linear association

The Pearson correlation captures the strength of the *linear* relationship and *only* the linear relationship. Two variables can be deterministically, perfectly related and still have a Pearson correlation of zero, provided the relationship is non-monotonic. The standard example is a variable X symmetric about zero and $Y = X^2$: every value of Y is exactly determined by X, yet because the upward slope on the right is mirrored by the downward slope on the left, the deviations cancel in the covariance and $\rho = 0$. A correlation of zero therefore does *not* mean "no relationship"; it means "no *linear* relationship." There may be a strong curved, cyclical, or threshold relationship that correlation is structurally blind to. This is the deeper reason that one must always *plot* the data, as urged in Chapter 2: a scatterplot reveals nonlinear structure that a single correlation coefficient hides. Datasets with wildly different shapes — a clean line, a curve, a blob with one outlier — can share an identical correlation, so the number must never be trusted without the picture.

9.3.2 Correlation does not imply causation

The most consequential warning in applied statistics is that *correlation does not imply causation*. That two variables move together tells you nothing, by itself, about whether one *causes* the other. There are always several explanations for an observed correlation, and the causal

one is merely a candidate among them. It may be that X causes Y; it may equally be that Y causes X (*reverse causation*); it may be that some third variable Z influences both, generating a correlation between them while neither affects the other (*confounding*); or the correlation may be a *coincidence*, a spurious pattern thrown up by chance, especially when many possible pairs of variables are screened and only the correlated ones reported. The confounding case is the most insidious and the most common in real data: a lurking variable drives two quantities in tandem, and a naive analysis mistakes their shared response for a direct link. Establishing causation requires more than correlation — ideally a controlled experiment that intervenes on the supposed cause while holding other factors fixed, or, where experiments are impossible, careful designs and assumptions that rule out the confounding and reverse-causation alternatives. The entire edifice of regression that follows can quantify and adjust for relationships among variables, but it is essential to understand from the outset that no amount of regression machinery converts an observed association into a demonstrated cause; the causal interpretation always rests on assumptions *outside* the data.

9.4 Rank correlation

The Pearson correlation's restriction to linear relationships, and its sensitivity to outliers (inherited from the means and standard deviations it is built on), motivate a robust alternative. The **Spearman rank correlation** is simply the Pearson correlation computed on the *ranks* of the data rather than their raw values: each observation is replaced by its position in the sorted order, and the ordinary correlation of these ranks is taken. This simple change has two valuable consequences. Because it depends only on order, the Spearman correlation measures the strength of any *monotonic* relationship — any relationship in which Y consistently increases (or consistently decreases) with X — not merely a linear one, so it detects curved-but-monotonic associations that the Pearson correlation underestimates. And because ranks are unaffected by the magnitude of extreme values, it is robust to outliers in the same way the median is. When a relationship is monotonic but nonlinear, or when the data contain outliers that would distort the Pearson coefficient, the Spearman correlation is often the more honest summary. It stands to the Pearson correlation as the median and IQR stand to the mean and standard deviation: a rank-based, robust counterpart to a moment-based, efficient original.

9.5 From correlation to regression

Correlation measures the strength and direction of a linear association symmetrically — it treats X and Y on equal footing and yields a single number. Often, though, we want more: we want to *describe the relationship as a function*, to predict Y from X, to quantify how much Y changes per unit change in X, and to separate the influence of several variables acting at once. This is the task of regression. Correlation and regression are intimately connected — in the simple two-variable case the regression slope is the correlation rescaled by the ratio of the standard deviations, and the squared correlation will turn out to be exactly the fraction of variance the regression explains — but regression goes further by giving an asymmetric, predictive, and extensible model of the relationship. The next three chapters develop it in full, beginning with the simplest case of one predictor and ending with a careful account of everything that can go wrong.

9.6 Summary

Covariance measures whether two variables co-move, with an interpretable sign but a unit-dependent and therefore uninterpretable magnitude. Dividing by the two standard deviations

yields the Pearson correlation, a unitless measure bounded between -1 and $+1$ whose magnitude reports the strength of the *linear* association on a universal scale. Two cautions govern its use: it captures only linear relationships, so a zero correlation can coexist with a strong nonlinear one, and it never establishes causation, since confounding, reverse causation, and coincidence are always live alternatives. The Spearman rank correlation provides a robust, monotonic-sensitive alternative. Correlation summarizes association in a single symmetric number; regression, which builds directly on these ideas, turns association into a predictive functional model, and it is the subject to which we now turn.

Chapter 10

Simple Linear Regression

Regression is the most used tool in applied quantitative analysis, and simple linear regression — one outcome explained by one predictor — is where its logic is clearest. The goal is to move beyond the symmetric, single-number summary of correlation to an asymmetric, predictive model: a fitted line that lets us predict an outcome from a predictor, quantify how much the outcome changes per unit change in the predictor, and attach uncertainty to both. Everything in this chapter generalizes directly to the many-predictor case of Chapter 11, so the concepts are worth establishing carefully here.

10.1 The model

We posit a linear relationship between a predictor X and an outcome Y, corrupted by random error.

Definition 10.1 (Simple linear regression model)**.** The **simple linear regression model** expresses the outcome as a linear function of the predictor plus an error term,

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where β_0 is the *intercept*, β_1 is the *slope*, and ε_i is a random *error* capturing everything about Y_i not explained by X_i.

The two coefficients have direct interpretations. The slope β_1 is the central quantity of interest: it is the expected change in the outcome for a one-unit increase in the predictor. The intercept β_0 is the expected outcome when the predictor is zero, which is meaningful only when a predictor value of zero is itself meaningful and within the range of the data. The error term ε_i is essential conceptually: it acknowledges that the relationship is not exact, that real data scatter around the line because of measurement noise, omitted influences, and inherent randomness. The line $\beta_0 + \beta_1 X$ describes the *average* outcome at each predictor value; individual outcomes deviate from it by their errors.

The coefficients β_0 and β_1 are unknown population parameters, the truth we wish to learn. From a sample we compute *estimates* $\hat{\beta}_0$ and $\hat{\beta}_1$, which define the **fitted line** $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$. For each observation the fitted value is $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, and the **residual** $e_i = y_i - \hat{y}_i$ is the vertical gap between the observed point and the fitted line — the sample's estimate of the unobservable error ε_i. The distinction between the true errors ε_i and the observed residuals e_i is one to keep firmly in mind; the residuals are what we can see and will use, in Chapter 12, to check whether the assumptions about the errors are tenable.

10.2 Estimating the line: ordinary least squares

There are infinitely many lines one could draw through a scatter of points; we need a principle for choosing the best one. The dominant principle is **ordinary least squares (OLS)**: choose the line that minimizes the sum of squared residuals.

Definition 10.2 (Least-squares criterion)**.** The OLS estimates are the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the **residual sum of squares**,

$$\text{SSE} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2.$$

Squaring the residuals before summing serves the same purpose as in the variance of Chapter 2: it prevents positive and negative residuals from cancelling, and it penalizes large misses disproportionately. Minimizing this sum is a calculus problem. Taking partial derivatives of SSE with respect to $\hat{\beta}_0$ and $\hat{\beta}_1$, setting them to zero, and solving the resulting two equations (the *normal equations*) yields closed-form expressions for the estimates:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{s_{XY}}{s_X^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

These two formulas are the computational core of simple regression and reward close reading. The slope estimate is the sample covariance of X and Y divided by the variance of X — co-movement per unit of predictor spread. Rewriting it in terms of the correlation of Chapter 9 gives the illuminating identity

$$\hat{\beta}_1 = r\,\frac{s_Y}{s_X},$$

so the regression slope is the correlation rescaled by the ratio of the outcome's spread to the predictor's spread. Correlation and regression are thus two expressions of the same underlying association: correlation is the unitless, symmetric version; the slope is the same information rendered in the units of the problem and made directional. The intercept formula, meanwhile, shows that the fitted line always passes through the point of averages $(\bar{x}, \bar{y})$ — the line is anchored at the center of the data and tilted by the slope.

10.3 The classical assumptions

OLS produces a line for any data whatsoever — the formulas above always compute. But for that line to support valid *inference*, and for the estimates to have good properties, a set of assumptions about the error term must hold. These assumptions are the fine print of regression, and Chapter 12 is devoted entirely to what happens when they fail; here we state them and explain what each buys.

1. **Linearity.** The relationship between the predictor and the mean outcome is genuinely linear: $\mathbb{E}[\varepsilon_i] = 0$ at every value of X, so the line correctly captures the average. If the true relationship is curved, a fitted straight line is biased — systematically wrong — no matter how much data is collected.

2. **Independence.** The errors are independent of one another. This typically holds for cross-sectional data but is frequently violated in time series, where consecutive errors are correlated; that violation, autocorrelation, is a principal subject of Chapter 12.

3. **Homoskedasticity.** The errors have constant variance σ^2 across all values of the predictor — the scatter around the line is equally wide everywhere. When instead the spread grows or shrinks with X, the errors are *heteroskedastic*, the other principal subject of Chapter 12.

4. **Normality.** The errors are normally distributed. This assumption is not needed for the estimates themselves to be good, but it is what makes the t- and F-based inference below exact in small samples; for large samples the Central Limit Theorem makes inference approximately valid even without it.

A useful way to remember these is that the errors should be independent draws from a single normal distribution with mean zero and constant variance, the same at every predictor value. The first three assumptions concern the estimates and their standard errors; the fourth concerns only the exactness of small-sample inference.

10.3.1 The Gauss–Markov theorem

The assumptions earn their keep through a celebrated optimality result.

Theorem 10.1 (Gauss–Markov)**.** *Under the assumptions of linearity, zero-mean errors, constant error variance (homoskedasticity), and uncorrelated errors, the ordinary least squares estimators are the* ***Best Linear Unbiased Estimators (BLUE)****: among all estimators that are both linear in the outcome and unbiased, they have the smallest variance.*

The Gauss–Markov theorem is the deep justification for least squares. It says that, given these assumptions, no other unbiased estimator that is linear in the data can be more precise than OLS — least squares extracts the most information possible under those conditions. Notably, the theorem does *not* require normality; normality is needed only for the exact small-sample distribution of the test statistics, not for the optimality of the estimates. The theorem also makes transparent why violations matter: when homoskedasticity or uncorrelated errors fail, OLS remains unbiased but loses its optimality and, more practically, its standard error formulas become wrong — the central problem addressed in Chapter 12.

10.4 How well does the line fit? The decomposition of variance

Having fit a line, we want to know how much of the outcome's variation it accounts for. The answer comes from partitioning the total variation in Y into a part explained by the regression and a part left over. Define three sums of squares:

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST, total}} \;=\; \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR, explained}} \;+\; \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{SSE, residual}}.$$

The **total sum of squares** (SST) measures the total variation of the outcome about its mean — the variation we are trying to explain. The **regression sum of squares** (SSR) measures the variation captured by the fitted line, the spread of the fitted values about the mean. The **residual sum of squares** (SSE) measures the variation left unexplained, the spread of the points about the line. That these add up exactly — total equals explained plus residual — is the fundamental decomposition of regression, and it holds precisely because least squares makes the residuals uncorrelated with the fitted values.

This partition yields the single most reported measure of fit.

Definition 10.3 (Coefficient of determination)**.** The **coefficient of determination** R^2 is the fraction of the total variation in the outcome explained by the regression,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}, \qquad 0 \le R^2 \le 1.$$

An R^2 of 0.7 means the predictor accounts for 70% of the variance in the outcome, leaving 30% unexplained. In simple regression R^2 equals exactly the square of the correlation coefficient, $R^2 = r^2$, tying this measure of fit directly back to Chapter 9. Two cautions apply. A high R^2 indicates the line explains much of the variation but says nothing about whether the model is *correct* — a curved relationship can produce a high R^2 from a wrong straight line

— and a low R^2 does not mean the relationship is unreal or useless, only that the predictor leaves much unexplained, which is the norm for noisy real-world outcomes. R^2 measures explained variance, not correctness, and it should always be read alongside the diagnostic plots of Chapter 12.

The unexplained variation also gives the estimate of the error standard deviation, the **residual standard error**,

$$s = \sqrt{\frac{\text{SSE}}{n-2}},$$

where the divisor is $n-2$ because two parameters — the slope and intercept — were estimated from the data, costing two degrees of freedom (compared with the single degree lost in estimating one mean in Chapter 2). This s is the typical size of a residual, in the units of the outcome, and it feeds directly into the standard errors of the coefficients.

10.5 Inference on the coefficients

The estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are statistics computed from a sample, so they have sampling distributions, and the most important inferential question is whether the slope is distinguishable from zero — whether the predictor has any linear relationship with the outcome at all. Under the classical assumptions, the slope estimate is unbiased with standard error

$$\text{se}(\hat{\beta}_1) = \frac{s}{\sqrt{\sum_i (x_i - \bar{x})^2}}.$$

This expression repays attention: the slope is estimated more precisely (smaller standard error) when the residual scatter s is small, when the sample is large, and when the predictor values are themselves widely spread (large $\sum (x_i - \bar{x})^2$). The last point is easy to overlook but important — a predictor that barely varies cannot reveal its effect, so spread in X is informative.

To test the null hypothesis $H_0 : \beta_1 = 0$ of no linear relationship, we form the standardized statistic

$$t = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)},$$

which under the null follows a Student t distribution with $n-2$ degrees of freedom — the same testing logic as Chapter 8, with the t distribution arising for the same reason, that we have estimated the error variance. A large value of $|t|$, equivalently a small p-value, is evidence that the slope is genuinely nonzero. The corresponding confidence interval for the slope, $\hat{\beta}_1 \pm t_{\alpha/2,n-2}\,\text{se}(\hat{\beta}_1)$, gives the range of slope values consistent with the data and is, as always, more informative than the bare test. The same construction applies to the intercept, though it is usually of less interest.

For simple regression there is also an **overall F-test** of the same null, built from the variance decomposition by comparing explained to residual variance, $F = \text{SSR}/(\text{SSE}/(n-2))$. In the single-predictor case this F statistic is exactly the square of the slope's t statistic and carries identical information, so the two tests always agree. The F-test earns its independent importance only in Chapter 11, where there are several coefficients to test jointly.

10.6 Confidence intervals versus prediction intervals

A subtle but practically vital distinction arises when we use the fitted line to say something about the outcome at a particular predictor value x_0. There are two genuinely different questions, with two different and famously confused answers.

The first question asks for the *average* outcome at x_0 — the height of the true regression line there, $\beta_0 + \beta_1 x_0$. The uncertainty in this quantity comes only from the uncertainty in the estimated coefficients, and it is quantified by a **confidence interval for the mean response**, with standard error $s\sqrt{\frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_i (x_i-\bar{x})^2}}$.

The second question asks for the value of a *single new* outcome at x_0 — not the average, but one particular future observation. This carries the same uncertainty about the line *plus* the irreducible scatter of an individual point about the line, and it is quantified by a **prediction interval**, whose standard error $s\sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_i (x_i-\bar{x})^2}}$ contains an extra "1+" under the root. That extra term is the variance of a single error, σ^2, which no amount of data can shrink away — even if we knew the line exactly, individual outcomes would still scatter. Consequently the prediction interval is always wider than the confidence interval, and the gap does not vanish as the sample grows. Confusing the two leads to drastically overconfident statements about individual outcomes, mistaking precision about an average for precision about a case. Both intervals also widen as x_0 moves away from the center $\bar{x}$ of the data, formalizing the danger of **extrapolation**: predictions far outside the observed range of the predictor are inherently unreliable, both because the standard errors balloon and, more fundamentally, because there is no evidence that the linear relationship continues to hold out there at all.

10.7 Summary

Simple linear regression models an outcome as a linear function of a predictor plus random error, estimating the line by ordinary least squares — minimizing the sum of squared residuals, which yields the slope as covariance over predictor variance, equivalently the correlation rescaled by the ratio of standard deviations. Valid inference rests on four assumptions about the errors — linearity, independence, constant variance, and normality — under which the Gauss–Markov theorem certifies OLS as the best linear unbiased estimator. The total variation of the outcome partitions into explained and residual pieces, giving R^2 as the fraction explained (equal to r^2 here) and the residual standard error with its $n - 2$ degrees of freedom. The slope is tested against zero with a t statistic and bracketed by a confidence interval, while predictions about outcomes require the crucial distinction between a confidence interval for the mean response and a wider prediction interval for an individual observation, with extrapolation beyond the data a standing hazard. All of this extends, with matrix notation doing the bookkeeping, to the multiple-predictor models of the next chapter.

Chapter 11

Multiple Linear Regression

Real outcomes depend on more than one thing. Multiple linear regression extends the model of Chapter 10 to several predictors at once, and in doing so it gains a capability that simple regression lacks entirely: the ability to isolate the effect of each predictor while *holding the others fixed*. This is what allows regression to "control for" confounding variables, and it is the single most important reason the technique dominates applied quantitative analysis. The price of the extension is notational — the cleanest treatment uses matrices — and conceptual — several new issues arise, multicollinearity and model selection chief among them. This chapter develops all of it.

11.1 The model and its matrix form

Definition 11.1 (Multiple linear regression model)**.** With p predictors, the model is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \varepsilon_i,$$

where each β_j is the coefficient on predictor j and ε_i is the error.

Writing this out for all n observations produces a system that cries out for matrix notation. Collect the outcomes into a vector $\mathbf{y}$ of length n, the coefficients into a vector $\boldsymbol{\beta}$ of length $p+1$ (including the intercept), and the predictors into an $n \times (p+1)$ **design matrix** $\mathbf{X}$ whose first column is all ones (for the intercept) and whose remaining columns hold the predictor values. The entire model then collapses to a single compact equation,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

This is not merely shorthand; the matrix form is what makes the theory tractable for any number of predictors, and it is the form in which all serious computation and derivation is done. The simple regression of Chapter 10 is the special case $p = 1$, where $\mathbf{X}$ has just two columns.

11.2 Estimation by least squares

The least-squares principle is unchanged: choose $\hat{\boldsymbol{\beta}}$ to minimize the residual sum of squares, now written $\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2$. Minimizing this leads, exactly as before but now using matrix calculus, to the **normal equations** $\mathbf{X}^\top \mathbf{X}\,\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$, whose solution is the central formula of multiple regression:

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}.$$

This single expression generalizes the slope and intercept formulas of Chapter 10 to arbitrarily many predictors. The matrix $\mathbf{X}^\top\mathbf{X}$ is a $(p+1) \times (p+1)$ matrix capturing the variances and covariances among the predictors; its inverse is what disentangles their overlapping influences. The fitted values are $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}$, where

$$\mathbf{H} = \mathbf{X}\left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top$$

is called the **hat matrix** because it puts the "hat" on $\mathbf{y}$, mapping observed outcomes to fitted values. The hat matrix will return in Chapter 12, where its diagonal entries measure the *leverage* of each observation — how much pull each data point exerts on its own fitted value — a key diagnostic for influential points. Under the classical assumptions the covariance matrix of the estimates is $\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$, from whose diagonal the standard errors of the individual coefficients are read.

For the inverse $(\mathbf{X}^\top\mathbf{X})^{-1}$ to exist, the predictor columns must be *linearly independent*: no predictor may be an exact linear combination of the others. This is the assumption of **no perfect multicollinearity**, and when it fails — for instance if one includes both a quantity and a perfectly redundant rescaling of it — the matrix cannot be inverted and the coefficients are not even defined. The more common and more insidious case of *near*-collinearity is taken up below.

11.3 Interpreting coefficients: the partial effect

The interpretation of a coefficient in multiple regression differs crucially from the simple case, and this difference is the whole point of the method.

Definition 11.2 (Partial effect)**.** In multiple regression, the coefficient β_j is the expected change in the outcome for a one-unit increase in predictor X_j, **holding all other predictors constant**.

The phrase "holding all other predictors constant" — often rendered *ceteris paribus* — is what separates multiple from simple regression. A simple regression of Y on X_j alone measures the *total* association between them, including any part that runs through other variables correlated with X_j. The multiple-regression coefficient instead measures the effect of X_j *net of* the other predictors in the model: it answers what happens when X_j moves but the others are held fixed. This is precisely how regression "controls for" confounders. If two variables both influence an outcome and are themselves correlated, including both in the regression allows each coefficient to capture that variable's own contribution, stripped of the other's. The hope, in observational data, is that by including enough of the relevant confounders one can approximate the effect that an experiment would have isolated.

Two warnings temper this power, both echoing Chapter 9. First, the ceteris paribus interpretation is only as good as the set of predictors included: variables that influence the outcome and are correlated with X_j but are *left out* of the model contaminate $\hat{\beta}_j$, a problem (omitted-variable bias) returned to in Chapter 12. Second, and most importantly, the ability to "control for" variables statistically does *not* convert a regression coefficient into a causal effect. The coefficient is a partial association; calling it an effect still rests on assumptions outside the data — that the right variables are controlled for, that there is no reverse causation, that the model is correctly specified. Regression is an extraordinarily flexible tool for adjusting associations, but it does not manufacture causation, and the most common abuse of multiple regression is presenting an adjusted coefficient as though the adjustment had settled the causal question.

11.4 Assessing fit with many predictors

The variance decomposition and R^2 of Chapter 10 carry over unchanged in form, but R^2 acquires a serious defect when predictors are added. Because including any new predictor — even a column of pure noise — can only reduce the residual sum of squares (the new variable can always be given a nonzero coefficient if it helps even slightly by chance), R^2 *never decreases when a predictor is added* and almost always increases a little. It therefore cannot be

used to compare models with different numbers of predictors, since it mechanically rewards complexity. The remedy is to penalize the addition of predictors that do not earn their place.

Definition 11.3 (Adjusted R^2)**.** The **adjusted** R^2 corrects the ordinary R^2 for the number of predictors p and sample size n,

$$\bar{R}^2 = 1 - \frac{\mathrm{SSE}/(n-p-1)}{\mathrm{SST}/(n-1)} = 1 - (1-R^2)\,\frac{n-1}{n-p-1}.$$

Adjusted R^2 uses the residual variance with its proper degrees-of-freedom divisor $n-p-1$ rather than the raw sum of squares, so adding a useless predictor — which barely lowers SSE while lowering the divisor — actually *reduces* adjusted R^2. It can decrease, and that is exactly its value: it rises only when a new predictor improves the fit by more than would be expected from chance alone, making it a legitimate, if rough, tool for comparing models of different sizes. The residual standard error is correspondingly $s = \sqrt{\mathrm{SSE}/(n-p-1)}$, with $p+1$ degrees of freedom spent on the intercept and p slopes.

11.5 Testing coefficients: t-tests and F-tests

Multiple regression poses two distinct kinds of testing question, and they require different tools.

The first asks about a *single* coefficient: does predictor X_j have an effect, holding the others fixed? This is answered exactly as in Chapter 10, by a **t-test** on that coefficient, $t = \hat{\beta}_j / \mathrm{se}(\hat{\beta}_j)$, compared to a t distribution with $n-p-1$ degrees of freedom. The crucial point is that this now tests the *partial* effect — the contribution of X_j beyond what the other predictors already explain — so a variable can be significant alone yet insignificant in the presence of correlated predictors, and vice versa.

The second asks about the model *as a whole* or about a *group* of coefficients, and here the single-coefficient t-tests are inadequate, because running many of them invites the multiple-comparisons problem of Chapter 8 and because they cannot assess joint significance. The **overall F-test** addresses the global null hypothesis that *all* slope coefficients are simultaneously zero — that the predictors jointly explain nothing — using the statistic

$$F = \frac{\mathrm{SSR}/p}{\mathrm{SSE}/(n-p-1)},$$

which compares explained variance per predictor to residual variance per degree of freedom and follows an F distribution with p and $n-p-1$ degrees of freedom. A significant overall F-test says the model has *some* explanatory power, even if no single coefficient stands out. It is entirely possible, when predictors are correlated, for the overall F-test to be highly significant while no individual t-test is — a configuration that is itself a signature of multicollinearity.

Between these extremes lies the **partial (nested) F-test**, which compares a larger model to a smaller one nested inside it to ask whether a *subset* of predictors, added as a block, significantly improves the fit. Fitting both the full and the reduced model and comparing their residual sums of squares,

$$F = \frac{(\mathrm{SSE}_{\text{reduced}} - \mathrm{SSE}_{\text{full}})/q}{\mathrm{SSE}_{\text{full}}/(n-p-1)},$$

where q is the number of predictors dropped, tests whether the extra predictors collectively matter. This nested F-test is the principled way to decide whether a group of related variables — a set of regime indicators, a block of interaction terms — belongs in the model, and it is the workhorse of structured model comparison.

11.6 Multicollinearity

When predictors are highly correlated with one another, the regression suffers from **multicollinearity**, a condition that does not bias the coefficients but severely undermines their precision and interpretability. The intuition follows directly from the partial-effect interpretation: if two predictors move almost in lockstep, the data contain little information about the effect of one *holding the other fixed*, because the two are almost never observed varying independently. The regression cannot cleanly separate their contributions.

The consequences are characteristic. The standard errors of the affected coefficients become large, so individual t-tests turn insignificant even when the predictors jointly matter — the configuration of a significant overall F with no significant individual coefficients noted above. The coefficient estimates become unstable, swinging wildly in magnitude and even sign in response to small changes in the data or the inclusion of other variables. And the coefficients can take values that are individually implausible while still fitting well collectively. The standard diagnostic is the **variance inflation factor (VIF)**, computed for each predictor by regressing it on all the *other* predictors and forming

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 of that auxiliary regression. A VIF of one indicates the predictor is uncorrelated with the others; large values (a common rule of thumb flags VIFs above five or ten) indicate problematic collinearity, the name reflecting that the variance of $\hat{\beta}_j$ is inflated by exactly this factor relative to the uncorrelated case. Remedies include dropping one of a redundant pair, combining collinear predictors into a single index, or collecting data with more independent variation in the predictors; one must also recognize that if the collinear variables are merely controls rather than the focus, their imprecise individual coefficients may not matter for the model's purpose.

11.7 Model selection

With many candidate predictors, a central question is which to include. This is the problem of **model selection**, and it is governed by a fundamental tension. A model with more predictors fits the observed data better (lower SSE, higher R^2) but risks **overfitting** — capturing the idiosyncratic noise of this particular sample rather than the stable structure of the population, which causes it to predict new data poorly. A model with fewer predictors is more robust and interpretable but risks missing real structure. Good model selection navigates between underfitting and overfitting, and it is the applied face of the bias–variance tradeoff from Chapter 7: too few predictors gives a biased model, too many gives a high-variance one.

Several criteria operationalize this tradeoff by rewarding fit while penalizing complexity. Adjusted R^2, already introduced, is the simplest. More principled are the information criteria, which add an explicit penalty per parameter to a measure of fit: the **Akaike Information Criterion (AIC)** and the **Bayesian Information Criterion (BIC)** both take the form of (a term decreasing in fit) plus (a penalty increasing in the number of parameters k),

$$\mathrm{AIC} \approx n \ln(\mathrm{SSE}/n) + 2k, \qquad \mathrm{BIC} \approx n \ln(\mathrm{SSE}/n) + k \ln(n),$$

with lower values preferred. The two differ in how harshly they penalize complexity: BIC's penalty $k \ln n$ exceeds AIC's $2k$ for any realistic sample size, so BIC favors smaller models and is the more conservative, while AIC is more permissive and oriented toward predictive accuracy. Automated **stepwise** procedures add or remove predictors one at a time according

to such a criterion, but they must be used with caution: they can overfit, they do not explore all combinations, and the resulting p-values are distorted because the same data were used both to choose the model and to test it.

The deepest point about model selection is that fit to the data in hand is the wrong target. The honest test of a model is its performance on *new* data it was not fit to, which is why **out-of-sample validation** — holding back part of the data, or using cross-validation, to evaluate predictions on observations the model has never seen — is the gold standard. A model selected purely to maximize in-sample fit will flatter itself; only out-of-sample performance reveals whether it has captured real structure or merely memorized noise. This principle, that genuine validation requires data not used in fitting, is among the most important in all of applied statistics and the surest guard against the seductive overconfidence of a high in-sample R^2.

11.8 Summary

Multiple regression models an outcome as a linear combination of several predictors, compactly written $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and solved by the matrix least-squares formula $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$. Its defining feature is the partial-effect interpretation: each coefficient measures one predictor's influence holding the others fixed, which is how regression adjusts for confounders — though never, by itself, establishing causation. Because ordinary R^2 rewards complexity mechanically, adjusted R^2 and the information criteria AIC and BIC are needed to compare models of different sizes, and inference splits into t-tests for individual partial effects, an overall F-test for joint significance, and nested F-tests for blocks of predictors. Highly correlated predictors cause multicollinearity, inflating standard errors and destabilizing estimates, diagnosed by the variance inflation factor. Model selection balances underfitting against overfitting — the bias–variance tradeoff again — and its only honest arbiter is out-of-sample performance. All of this assumes the classical conditions on the errors hold; the final chapter confronts what to do when they do not.

Chapter 12

When the Assumptions Break

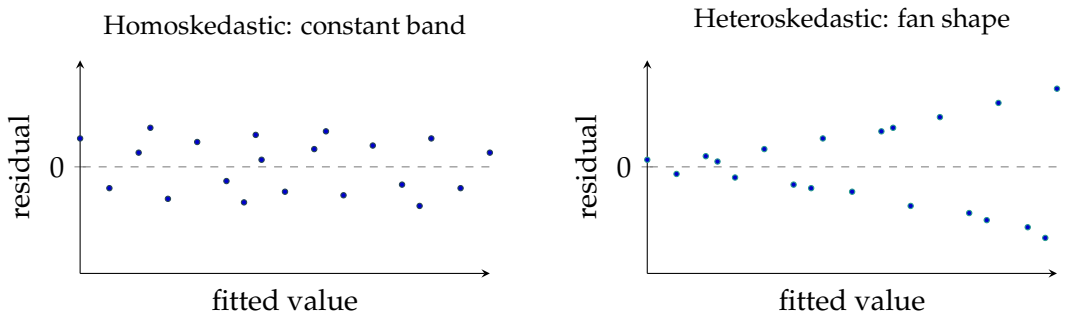
The regression theory of the last two chapters rested on a set of assumptions about the error term. In real data those assumptions are routinely violated, and the most valuable skill in applied regression is not fitting models but *diagnosing* them — recognizing when an assumption has failed, understanding what the failure does to the conclusions, and knowing how to respond. This closing chapter takes each major violation in turn. The unifying tool throughout is the **residuals**: since the true errors are unobservable, the residuals $e_i = y_i - \hat{y}_i$ are our only window onto the error term's behavior, and nearly every diagnostic is, at bottom, a way of looking at them.

12.1 Heteroskedasticity

The homoskedasticity assumption holds that the errors have constant variance across all values of the predictors. **Heteroskedasticity** is its violation: the spread of the errors changes systematically, often growing as a predictor or the fitted value grows. It is extremely common — larger quantities frequently exhibit larger absolute fluctuations — and in cross-sectional and financial data it is closer to the rule than the exception.

The consequences are specific and must be understood precisely, because heteroskedasticity is often misdiagnosed as a more serious problem than it is. The OLS coefficient estimates remain *unbiased* and *consistent* — heteroskedasticity does not push them toward wrong values. What it destroys is the validity of their *standard errors*. The usual standard-error formulas of Chapters 10 and 11 are derived under constant variance, and when that fails they are simply wrong, typically understated. Since every t-test, F-test, confidence interval, and p-value is built on those standard errors, all of the *inference* becomes unreliable: significance is overstated, intervals are too narrow, and one is led to false confidence. The estimates are fine; the error bars are not.

Detection proceeds first by eye and then by formal test. The primary graphical tool is a plot of the residuals against the fitted values: under homoskedasticity the residuals form a band of constant vertical width, while heteroskedasticity reveals itself as a systematic change in that width — most classically a fan or funnel shape, where the scatter widens as the fitted value increases. Formal tests put a number on this. The **Breusch–Pagan test** regresses the squared residuals on the predictors and checks whether they have any explanatory power — if the squared residuals can be predicted from the regressors, their variance is non-constant. The **White test** is a more general version that also includes squares and cross-products of the predictors, detecting more general forms of heteroskedasticity without assuming a particular pattern.



The remedies fall into two families. The most common and least disruptive is to keep the OLS estimates but replace the standard errors with **heteroskedasticity-consistent (robust) standard errors**, also called White or sandwich standard errors, which are valid whatever the pattern of the error variance and have become a near-default in much applied work. The alternative is **weighted least squares (WLS)**, which, when the form of the heteroskedasticity is known, weights each observation inversely to its error variance so that noisier observations count less; this restores efficiency rather than merely correcting the standard errors, but it requires knowing or modeling the variance structure. Sometimes a *transformation* of the outcome, such as taking logarithms, stabilizes the variance directly and is the cleanest fix of all when the heteroskedasticity arises from multiplicative scale effects — another reason, beyond those in Chapter 5, that log transformations are pervasive.

12.2 Autocorrelation

The independence assumption holds that the errors are uncorrelated across observations. **Autocorrelation** (or serial correlation) is its violation: the error in one observation is correlated with the errors in others. It is the characteristic problem of *time-series* data, where consecutive observations are linked through time and a shock in one period tends to persist into the next, so that the assumption of independent errors — reasonable for a cross-section of unrelated units — is typically false for a sequence of observations on the same process over time.

The consequences parallel those of heteroskedasticity, and the parallel is worth drawing because the two are the twin failures of the error-structure assumptions. As with heteroskedasticity, OLS remains *unbiased* and *consistent*; the damage is again to the *standard errors*, which are wrong and, in the most common case of *positive* autocorrelation, badly understated. Positive autocorrelation means successive errors tend to share sign, so the data carry less independent information than their count suggests — runs of observations move together rather than each contributing afresh. OLS, blind to this, treats every observation as independent and thereby overstates the effective sample size, producing standard errors that are too small, t-statistics that are too large, and significance that is spurious. In time-series regression, undetected positive autocorrelation is a leading cause of finding "significant" relationships that are nothing of the kind.

Detection again combines plots and tests. Plotting the residuals *in time order* is the first step: independent errors look like noise, while autocorrelated errors show patterns — long runs above and below zero for positive autocorrelation, or rapid alternation for negative. The classic formal test is the **Durbin–Watson statistic**, which measures the correlation between consecutive residuals and takes values near 2 under no autocorrelation, below 2 for positive autocorrelation, and above 2 for negative. The Durbin–Watson test detects only first-order (adjacent) autocorrelation; the more general **Breusch–Godfrey test** regresses the residuals on their own lags and the predictors, detecting higher-order autocorrelation patterns that Durbin–Watson misses.

The remedies mirror the heteroskedasticity case. The standard-error-only fix is to use **HAC (heteroskedasticity- and autocorrelation-consistent) standard errors**, of which the **Newey–West** estimator is the best known; these correct the standard errors for both autocorrelation and heteroskedasticity at once while leaving the OLS estimates in place, and they are the routine choice in applied time-series work. The more ambitious approach is to *model the dynamics* directly — including lagged variables, or specifying a model for the correlated error structure and estimating by generalized least squares — which addresses the source rather than patching the symptom and can recover efficiency. The choice between patching the standard errors and modeling the dynamics is a recurring judgment in time-series analysis.

12.3 Non-normality of the errors

The normality assumption holds that the errors follow a normal distribution. Its status is different from the previous two violations, and the difference is reassuring. Non-normality does *not* bias the estimates, and crucially it does not invalidate inference in *large* samples, because the Central Limit Theorem of Chapter 6 ensures that the sampling distribution of the coefficient estimates is approximately normal regardless of the error distribution, provided the sample is large enough and the error variance is finite. Normality of the errors matters chiefly for the *exactness* of t- and F-based inference in *small* samples; with abundant data, moderate non-normality is largely harmless.

The situation that genuinely warrants concern is not mild non-normality but *heavy tails*. The thin-tailed normal is a poor model for errors that occasionally take very large values, and the connection to Chapter 2's discussion of excess kurtosis is direct: heavy-tailed errors produce occasional large residuals, which, through the squaring inherent in least squares, exert outsized influence on the fit and inflate the variance of the estimates. Detection uses the residuals: a **Q–Q plot** of the residuals against the normal is the standard graphical check — the heavy-tail signature of points curving away from the line at both ends, introduced in Chapter 2, is exactly what to look for — supplemented by histograms and by formal normality tests such as the Jarque–Bera test (which examines sample skewness and kurtosis) or the Shapiro–Wilk test. Remedies, when needed, include transforming the outcome to tame skewness, using *robust regression* methods that down-weight extreme observations rather than letting squared residuals dominate, or using the *bootstrap* — resampling the data to approximate the sampling distribution empirically — to obtain inference that does not rely on the normality assumption at all.

12.4 Outliers, leverage, and influence

Individual observations can distort a regression out of all proportion to their number, and disentangling how requires three distinct concepts that are routinely and damagingly conflated.

An **outlier** is an observation with a large residual — one that the model fits poorly, lying far from the fitted line in the outcome direction. **Leverage** is a different and independent property: it measures how extreme an observation is in the space of the *predictors*, how unusual its X-values are. Leverage is quantified by the diagonal entries of the hat matrix from Chapter 11, which record how much each observation's own outcome determines its own fitted value; a high-leverage point sits far from the center of the predictor cloud and therefore has the *potential* to pull the line strongly toward itself. The two properties are orthogonal — a point can have a large residual but little leverage (unusual outcome at a typical predictor value, which barely moves the line), or high leverage but a small residual (unusual predictor value that the line happens to fit well).

What actually matters is the combination of the two, captured by the third concept.

Definition 12.1 (Influence)**.** An **influential observation** is one whose removal would substantially change the fitted regression. Influence combines outlyingness and leverage: a point is influential when it is both far from the fitted line *and* has high leverage.

The reason the distinction matters is that influence, not mere outlyingness, is the real danger. A high-leverage point that is also an outlier can single-handedly tilt the entire regression, dragging the line toward itself and corrupting every coefficient — and because it pulls the line toward itself, its own residual may end up small, hiding it from a naive search for large residuals. The standard summary measure is **Cook's distance**, which for each observation quantifies how much all the fitted values change when that observation is deleted, neatly combining the size of its residual with its leverage into a single influence score; observations with conspicuously large Cook's distance warrant scrutiny. The correct response to an influential point is emphatically *not* to delete it reflexively. It is to *investigate* it: an influential observation may be a data error, in which case it should be corrected or removed, but it may equally be a genuine and important extreme case carrying real information, in which case discarding it would be discarding the most informative data point and biasing the analysis toward a comforting but false regularity. When influential points are genuine but troubling, robust regression methods that limit any single observation's pull offer a principled alternative to the crude binary of keep-or-delete.

12.5 Specification error

The most fundamental assumption of all is that the model is *correctly specified* — that the linear form is right and that the relevant variables are included. Violations of this assumption, collectively **specification errors**, are the most serious because they bias the coefficients themselves, and no correction to the standard errors can repair a model that is estimating the wrong thing.

The gravest case is **omitted-variable bias**. If a variable that genuinely affects the outcome is left out of the model, *and* that omitted variable is correlated with an included predictor, then the included predictor's coefficient is biased — it absorbs part of the omitted variable's effect, because the regression has no way to attribute that effect correctly. This is the formal mechanism behind the correlation-is-not-causation warnings of Chapters 9 and 11: an apparent effect of X on Y may in truth be the effect of a lurking confounder correlated with X, and omitting that confounder loads its influence onto $\hat{\beta}_X$. The direction of the bias can even be reasoned out from the sign of the omitted variable's effect and the sign of its correlation with the included predictor, so the distortion is not random but systematic. The opposite error, including *irrelevant* variables, is far less damaging — it does not bias the coefficients — but it inflates their variances and squanders degrees of freedom, the overfitting concern of Chapter 11. The asymmetry is worth internalizing: omitting a relevant variable biases, while including an irrelevant one merely adds noise, which is one argument for erring toward including plausible controls — tempered by the multicollinearity and overfitting costs of including too many.

A second form of specification error is **wrong functional form**: imposing a straight line on a relationship that is genuinely curved. The linearity assumption then fails, and the fitted model is biased everywhere, systematically over- and under-predicting in different regions of the data. This failure is visible in the residuals — a plot of residuals against a predictor or against fitted values will show a curved pattern rather than featureless scatter when the functional form is wrong — and it is detected formally by tests such as Ramsey's RESET test, which checks whether powers of the fitted values have explanatory power that the linear model missed. The remedy is to enrich the form: add polynomial terms, transform variables, or include interaction terms that let one predictor's effect depend on another.

Underlying both is the deepest specification problem, **endogeneity**: the situation in which

a predictor is correlated with the error term, which is precisely what omitted confounders, simultaneity (the outcome and predictor influencing each other), and measurement error in the predictors all produce. Endogeneity is the one violation that breaks OLS at the root, rendering the estimates not merely inefficient but *biased and inconsistent* — wrong even with infinite data. It is the reason that observational regression cannot, on its own, establish causation, and addressing it requires going beyond OLS to methods such as instrumental variables that exploit external sources of variation in the predictor uncorrelated with the error. A full treatment belongs to econometrics, but the conceptual point is the fitting end of this book: the validity of a regression depends ultimately not on the arithmetic of least squares, which always produces an answer, but on whether the assumptions connecting that answer to reality actually hold.

12.6 The diagnostic mindset

The thread tying this chapter together is a habit of mind. A fitted regression is the beginning of the analysis, not the end. Every model should be interrogated: plot the residuals against fitted values and against each predictor to check linearity and constant variance; plot them in time order if the data are sequential to check for autocorrelation; examine their distribution against the normal to check the tails; and look for influential observations through leverage and Cook's distance. Each of the violations in this chapter has a characteristic residual signature, which is why the residuals are the master diagnostic, and each has both a graphical check and a formal test, the two complementing rather than replacing each other — the plot reveals the nature of a problem, the test gauges its strength. Knowing the remedies matters less than knowing to look, because the gravest errors in applied regression come not from choosing the wrong fix but from never noticing that anything was wrong. The arithmetic of least squares is the easy part; the judgment to know when its assumptions hold, and the discipline to check, is the whole of the craft.

12.7 Summary

When the regression assumptions break, the failures sort into a clear taxonomy with distinct consequences. Heteroskedasticity and autocorrelation leave the coefficients unbiased but invalidate the standard errors — the former from non-constant error variance, detected by residual-versus-fitted plots and the Breusch–Pagan and White tests and remedied by robust standard errors or weighted least squares; the latter from correlated errors in time-series data, detected by the Durbin–Watson and Breusch–Godfrey tests and remedied by Newey–West standard errors or dynamic modeling. Non-normality threatens only small-sample inference, since the Central Limit Theorem rescues large samples, with heavy tails the real concern, checked by a residual Q–Q plot. Individual observations distort the fit through the interplay of outlyingness and leverage, with influence — measured by Cook's distance — the quantity that matters and investigation, not deletion, the correct response. Specification errors are the gravest, because omitted variables, wrong functional form, and endogeneity bias the coefficients themselves, breaking the link between the regression and reality that no standard-error correction can restore. Across all of them, the residuals are the window and the diagnostic habit is the skill: a model is to be checked, not merely fitted. That discipline — reasoning from data, under uncertainty, while honestly testing the assumptions on which the reasoning depends — is the foundation this book set out to build, and the place from which all further quantitative work proceeds.

Part V

Further Methods

Chapter 13

Analysis of Variance

Chapter 8 gave the two-sample t-test for comparing the means of two groups. A natural and frequent need is to compare the means of *several* groups at once — three production methods, four regimes, five regions. One might imagine running a t-test on every pair, but this founders on the multiple-comparisons problem of Chapter 8: with many pairwise tests, the chance of at least one false positive balloons, and the nominal significance level becomes meaningless. **Analysis of variance (ANOVA)** solves this by testing all the group means simultaneously with a single test, and it does so, paradoxically, by analyzing variances. ANOVA is also the conceptual link between the t-tests of Part III and the regression of Part IV, because it is, underneath, a regression on group-indicator variables.

13.1 One-way ANOVA

Suppose we have k groups and wish to test whether they share a common mean.

Definition 13.1 (One-way ANOVA hypotheses)**.** For k groups with population means $\mu_1, \dots, \mu_k$, one-way ANOVA tests

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{against} \quad H_1 : \text{at least one mean differs.}$$

The alternative is deliberately weak — it asserts only that the means are not all equal, not which ones differ or in what direction. A significant ANOVA tells you that *something* is going on among the groups; pinning down *what* requires the follow-up comparisons described below.

13.1.1 The partition of variance

The name "analysis of variance" reflects the central idea: the total variation in the data is partitioned into a part *between* the groups and a part *within* them, and the test compares these two. Let N be the total number of observations across all groups, $\bar{y}$ the grand mean of all the data, and $\bar{y}_j$ the mean of group j. The total sum of squares decomposes exactly as

$$\underbrace{\sum_j \sum_i (y_{ij} - \bar{y})^2}_{\text{SST, total}} = \underbrace{\sum_j n_j (\bar{y}_j - \bar{y})^2}_{\text{SSB, between groups}} + \underbrace{\sum_j \sum_i (y_{ij} - \bar{y}_j)^2}_{\text{SSW, within groups}}.$$

The **between-group sum of squares** (SSB) measures how far the group means scatter from the grand mean — the variation *explained* by group membership. The **within-group sum of squares** (SSW) measures how far individual observations scatter from their own group's mean — the variation *not* explained by group, the residual noise. This is precisely the regression decomposition of Chapter 10 with "group" playing the role of the predictor, which is the formal sense in which ANOVA is regression.

13.1.2 The F-statistic

Dividing each sum of squares by its degrees of freedom yields the *mean squares*, and the test is their ratio. The between-group mean square is $\text{MSB} = \text{SSB}/(k-1)$, on $k-1$ degrees of freedom (one lost in computing the grand mean from k group means), and the within-group mean square is $\text{MSW} = \text{SSW}/(N-k)$, on $N-k$ degrees of freedom. The test statistic is

$$F = \frac{\text{MSB}}{\text{MSW}},$$

which under the null hypothesis follows an F distribution with $k-1$ and $N-k$ degrees of freedom (Chapter 5). The logic is transparent: if the group means are truly equal, the between-group variation reflects only the same random noise as the within-group variation, so the ratio is near one. If the group means genuinely differ, the between-group variation is inflated by the real differences, pushing F above one. A large F, equivalently a small p-value, is evidence that the means are not all equal. The within-group mean square MSW is, notably, just the pooled estimate of the common error variance σ^2, tying the denominator back to the residual variance of regression.

Example 13.1 (Reading an ANOVA)**.** Suppose three groups of ten observations each ($k = 3$, $N = 30$) give a between-group sum of squares of $SSB = 48$ and a within-group sum of squares of $SSW = 108$. Then $\text{MSB} = 48/(3-1) = 24$ and $\text{MSW} = 108/(30-3) = 4$, so $F = 24/4 = 6.0$ on 2 and 27 degrees of freedom. Comparing to the $F_{2,27}$ distribution, this exceeds the 5% critical value (about 3.35), so we reject the null and conclude the group means are not all equal. The fraction of variation explained by group is $SSB/SST = 48/156 \approx 0.31$, the ANOVA analogue of R^2.

13.1.3 Assumptions and post-hoc comparisons

ANOVA rests on the same assumptions as the regression it secretly is: the observations are independent, the within-group errors are normally distributed, and — importantly — the groups share a common variance (homoskedasticity across groups). Moderate departures from normality are forgiven in large samples by the Central Limit Theorem, but markedly unequal group variances distort the test, and when variances differ substantially, modifications such as Welch's ANOVA are preferred.

A significant F-test establishes only that the means are not all equal, not which differ. Identifying the specific differences requires **post-hoc multiple comparisons**, which compare pairs of groups while controlling the overall error rate inflated by making many comparisons. **Tukey's honestly significant difference** procedure is the standard method designed specifically for all pairwise comparisons, controlling the probability of any false positive across the whole family; the **Bonferroni correction**, which simply divides the significance level by the number of comparisons, is a simpler and more conservative general-purpose alternative. The two-stage logic — an omnibus F-test to establish that differences exist, then controlled pairwise comparisons to locate them — is the disciplined way to compare many groups without succumbing to the multiple-comparisons trap.

13.2 Two-way ANOVA

Often observations are classified by *two* factors at once — say, a method and a location — and we want to assess the effect of each, and whether they interact. **Two-way ANOVA** extends the partition to two factors plus their interaction, decomposing the total variation into a part due to the first factor (its *main effect*), a part due to the second factor (its main effect), a part due to

their **interaction**, and the residual within-cell variation. Each component yields its own F-test against the residual mean square.

The genuinely new concept is the interaction, and it is an important one. The *main effect* of a factor is its average effect across the levels of the other factor. An **interaction** is present when the effect of one factor *depends on the level of the other* — when, for instance, one method is best in one location but worst in another. Without an interaction, the factors' effects simply add; with one, they cannot be understood separately, because each factor's influence is contingent on the other's setting. Testing for interaction is frequently the main scientific point of a two-way design, since interactions reveal that the world is not additively separable, and interpreting main effects in the presence of a strong interaction is misleading — the average effect can obscure that the factor helps in some conditions and hurts in others. This is the same idea as an interaction term in a multiple regression (Chapter 11), again underscoring that ANOVA and regression are one framework viewed through two vocabularies.

13.3 Summary

Analysis of variance compares the means of several groups with a single test, sidestepping the multiple-comparisons inflation of repeated pairwise t-tests. It works by partitioning total variation into between-group and within-group components and comparing them through an F-statistic: large between-group variation relative to within-group noise signals that the means differ. Because it rests on independence, normality, and equal variances, it is the same model as regression on group indicators, with the within-group mean square estimating the common error variance. A significant omnibus test is followed by controlled post-hoc comparisons — Tukey or Bonferroni — to locate the differences. Two-way ANOVA adds a second factor and, crucially, the interaction between factors, which captures the non-additive situation in which one factor's effect depends on the other's level. With ANOVA understood as regression in disguise, the unity of the linear-model framework across Parts III, IV, and V comes fully into view.

Chapter 14

Nonparametric Methods

The inferential procedures of Part III lean on distributional assumptions — most often that the data, or the errors, are normal. When those assumptions are badly violated, when the sample is too small for the Central Limit Theorem to rescue them, or when the data are ordinal rather than numerical, the standard tests can mislead. **Nonparametric methods** are designed for exactly these situations. They make far weaker assumptions about the underlying distribution — typically nothing about its shape — and they achieve this robustness by working with the *ranks* of the data rather than the raw values, inheriting the same insensitivity to outliers and to distributional shape that made the median and the Spearman correlation robust in earlier chapters.

14.1 Why and when to go nonparametric

The appeal of nonparametric methods is that their validity does not depend on the data being normal or on any particular distributional form, so they remain trustworthy precisely where the parametric tests become questionable: heavy-tailed data, conspicuous outliers, small samples, and ordinal measurements for which means are not even meaningful. The cost is a loss of *power* when the parametric assumptions actually do hold — when the data really are normal, a rank-based test extracts somewhat less information than the corresponding t-test or ANOVA, because reducing values to ranks discards their magnitudes. The tradeoff is therefore a familiar one: nonparametric tests sacrifice some efficiency under ideal conditions in exchange for robustness under adverse ones. The loss of power is often surprisingly small, however, and when the parametric assumptions are doubtful the nonparametric test is not merely safer but more honest. Most parametric tests have a direct nonparametric counterpart, and the chapter is organized around those pairings.

14.2 The sign test

The simplest nonparametric test concerns the *median*. The **sign test** assesses whether the median of a population equals a hypothesized value by counting how many observations fall above it versus below. Under the null hypothesis that the value is the true median, each observation is equally likely to land above or below, so the number above follows a binomial distribution with success probability one-half (Chapter 5) — and the test reduces to checking whether the observed count of positive signs is consistent with that binomial. It uses only the *direction* of each deviation, not its size, which makes it maximally robust but also relatively low in power, since it throws away the magnitudes entirely. The sign test extends naturally to paired data by applying it to the signs of the within-pair differences, giving a distribution-free counterpart to the paired t-test that assumes essentially nothing about the differences beyond a common median.

14.3 The Wilcoxon signed-rank test

The sign test's weakness is that it ignores magnitudes; the **Wilcoxon signed-rank test** recovers some of that information while remaining distribution-free. For one-sample or paired data, it ranks the *absolute* values of the deviations from the hypothesized median, then sums the ranks of the positive deviations and compares that sum to its null distribution. By using the ranks of the magnitudes rather than just their signs, it accounts for the fact that a large deviation is stronger evidence than a small one, and it is correspondingly more powerful than the sign test. It assumes only that the distribution of the differences is symmetric, a far weaker condition than normality, and it is the standard distribution-free alternative to the one-sample and paired t-tests. When paired data show heavy tails or outliers that would distort a paired t-test, the Wilcoxon signed-rank test is the natural choice.

14.4 The Mann–Whitney U test

For comparing *two independent* groups — the nonparametric counterpart of the two-sample t-test — the standard tool is the **Mann–Whitney U test**, equivalently the Wilcoxon rank-sum test. Its mechanism is elegant: pool all the observations from both groups, rank them together from smallest to largest, and then examine the sum of the ranks belonging to each group. If the two groups are drawn from the same distribution, the ranks should be thoroughly intermingled and each group's rank sum should be close to what random allocation would give; if one group tends to produce larger values, its observations will cluster at the high ranks and its rank sum will be conspicuously large. The test statistic measures this rank-sum imbalance against its null distribution. Because it depends only on the ordering of the combined data, the Mann–Whitney test makes no assumption about the shape of the distributions and is robust to outliers, while still responding to systematic differences in level between the groups. It is the workhorse nonparametric two-group comparison and the test to reach for when the two-sample t-test's normality assumption is untenable.

14.5 The Kruskal–Wallis test

Extending the Mann–Whitney idea from two groups to many gives the **Kruskal–Wallis test**, the nonparametric counterpart of the one-way ANOVA of Chapter 13. As before, all observations across the k groups are pooled and ranked together, and the test examines whether the average ranks differ across groups more than random allocation would produce. Its statistic is built from the group rank-sums and, under the null that all groups share a common distribution, approximately follows a chi-squared distribution with $k - 1$ degrees of freedom (Chapter 5). The Kruskal–Wallis test thus tests the same broad hypothesis as one-way ANOVA — that the groups do not all have the same central tendency — without requiring the normality and equal-variance assumptions that ANOVA needs, and as with ANOVA a significant result is followed by post-hoc rank-based comparisons to locate the specific differences. It is the method of choice for comparing several groups when the data are ordinal or markedly non-normal.

14.6 The place of rank methods

The nonparametric tests form a coherent family paralleling the parametric one: the sign and signed-rank tests stand in for the one-sample and paired t-tests, the Mann–Whitney test for the two-sample t-test, and the Kruskal–Wallis test for one-way ANOVA, while the Spearman rank correlation of Chapter 9 is the rank-based counterpart of the Pearson correlation. What unifies

them is the move to ranks, which severs the tests' dependence on distributional shape and confers robustness to outliers, at the price of some efficiency when the parametric assumptions genuinely hold. A closely related and increasingly important approach, *permutation testing*, achieves distribution-free inference by a different route — directly constructing the null distribution by reshuffling the data — and is taken up among the resampling methods of Chapter 18. The practical guidance is straightforward: when the assumptions of a parametric test are well supported, prefer it for its greater power; when they are doubtful, and especially with small samples, heavy tails, or ordinal data, the corresponding nonparametric test trades a little power for a great deal of safety.

14.7 Summary

Nonparametric methods provide inference that does not depend on the distributional assumptions underpinning the t-tests and ANOVA of Part III, achieving robustness to non-normality and outliers by analyzing ranks rather than raw values. The sign test and the more powerful Wilcoxon signed-rank test handle one-sample and paired comparisons about a median; the Mann–Whitney U test compares two independent groups; and the Kruskal–Wallis test compares several, each serving as the distribution-free counterpart of a familiar parametric procedure. The methods cost some power when normality actually holds but become the more trustworthy choice when it does not, particularly for small or ordinal samples. Together with the rank correlation of Chapter 9 and the permutation tests of Chapter 18, they round out the inferential toolkit with procedures that remain valid when the convenient assumption of normality cannot be sustained.

Chapter 15

Time Series Analysis

Much quantitative data arrives as a *time series*: a single quantity observed repeatedly over time, each value indexed by when it occurred. Prices, rates, demand, output, and almost everything one tracks sequentially are time series, and they violate the assumption that has quietly underpinned most of this book — that observations are independent. In a time series, consecutive observations are typically dependent: today's value carries information about tomorrow's. This dependence is not a nuisance to be corrected but the central feature to be modeled, because it is exactly what makes forecasting possible. This chapter develops the core ideas, which also illuminate the autocorrelation problem encountered in regression diagnostics (Chapter 12).

15.1 What makes time series different

A time series is an ordered sequence, and the ordering matters: shuffling the observations destroys their meaning, unlike the exchangeable samples of earlier chapters. The dependence between nearby observations means the effective amount of information is less than the raw count suggests, the same point that made undetected autocorrelation so dangerous in regression. It is conventional to think of a series as composed of several superimposed components: a long-run **trend** (persistent upward or downward movement), a **seasonal** component (regular fluctuations tied to the calendar), a **cyclical** component (longer, irregular swings), and an **irregular** or random component (the unpredictable remainder). Much of classical time-series analysis is the project of separating these components, modeling the predictable ones, and capturing the dependence in the remainder.

15.2 Stationarity

The single most important concept in time-series analysis is **stationarity**, because the entire modeling apparatus assumes it. Intuitively, a series is stationary if its statistical properties do not change over time — it has no trend, its variance is stable, and its pattern of dependence is the same in every period.

Definition 15.1 (Weak stationarity)**.** A time series is **(weakly) stationary** if its mean is constant over time, its variance is constant over time, and the covariance between two observations depends only on the *lag* between them — the time gap — and not on *when* they occur.

Stationarity matters because it is what makes the past a reliable guide to the future. If a series' properties drift over time, then statistics estimated from past data — its mean, its variance, its dependence structure — do not describe the future, and forecasting from them is unfounded. A stationary series, by contrast, has stable properties that can be estimated from history and projected forward. Most real series of interest are *not* stationary as observed — they trend, or their variance grows — so a recurring first task is to *transform* a nonstationary series into a stationary one, most commonly by *differencing*: replacing each value by its change from the previous period, which removes a trend, and sometimes differencing again or taking

logarithms to stabilize a growing variance. The number of differences needed to achieve stationarity is a key descriptor of a series.

15.3 Autocorrelation and the dependence structure

The dependence within a stationary series is summarized by how each observation correlates with earlier ones. The **autocovariance** at lag k is the covariance between observations k periods apart, and standardizing it gives the **autocorrelation function (ACF)**: the correlation between the series and itself shifted by k periods, as a function of the lag k. The ACF is the time-series analogue of the correlation of Chapter 9, applied to a series and its own past, and it is the primary tool for diagnosing the dependence structure. A complementary summary, the **partial autocorrelation function (PACF)**, gives the correlation between observations k periods apart *after removing* the influence of the intervening observations — the direct lag-k relationship net of the shorter lags, much as a partial regression coefficient isolates one predictor's effect in Chapter 11. The ACF and PACF together fingerprint a series' dependence and are the basis for choosing a model.

The simplest stationary series is **white noise**: a sequence of uncorrelated observations with constant mean and variance, the time-series embodiment of pure randomness with no exploitable structure. Its ACF is zero at every nonzero lag. White noise is the benchmark of unpredictability, and a central goal of modeling is to capture all the structure in a series so that what remains — the residual series — is white noise. If the residuals from a fitted time-series model are not white noise, exploitable structure has been left on the table.

15.4 The random walk and unit roots

A particular nonstationary series deserves special attention because it is both ubiquitous and treacherous: the **random walk**, in which each value equals the previous value plus an independent random shock. The random walk is nonstationary — its variance grows without bound over time, since the shocks accumulate rather than wash out — and its best forecast of the next value is simply the current value, the increments being unpredictable. Many financial price series behave approximately as random walks, which is the formal content of the claim that price *changes* are close to unpredictable even though the price *level* is highly persistent.

The random walk is the leading example of a series with a **unit root**, a technical condition characterizing the boundary between stationary persistence and outright nonstationarity. Distinguishing a series with a unit root (which must be differenced to become stationary) from a merely persistent but stationary series is both important and statistically delicate, and it is the purpose of **unit root tests**, of which the **Augmented Dickey–Fuller test** is the standard. The stakes are high because of the phenomenon of **spurious regression**: regressing one nonstationary series on another unrelated nonstationary series tends to produce impressively significant results — high R^2, large t-statistics — that are entirely illusory, an artifact of both series trending rather than any real relationship. This is one of the most important cautions in applied time-series work, and it is why establishing stationarity, by differencing or by testing for unit roots, is a prerequisite to regression on time-series data, not an afterthought.

15.5 ARMA and ARIMA models

The workhorse models for stationary series build the current value out of its own past and the past shocks, in two complementary ways.

An **autoregressive (AR)** model of order p, written AR(p), expresses the current value as a linear combination of the previous p values plus a fresh shock — the series regressed on

its own recent history. Autoregression captures persistence and momentum: a high value tends to be followed by high values to the extent the autoregressive coefficients are positive. A **moving-average (MA)** model of order q, written MA(q), instead expresses the current value as a linear combination of the current and previous q *shocks* — the series as a weighted memory of recent disturbances. The two mechanisms are different: autoregression carries forward past *values*, moving-average carries forward past *shocks*, and they leave distinct signatures in the ACF and PACF, which is how one identifies which model a series calls for.

Combining them gives the **ARMA(p, q)** model, which includes both autoregressive and moving-average terms and can parsimoniously capture a wide range of stationary dependence structures. For nonstationary series, the model is applied not to the raw series but to its differenced, stationary form, giving the **ARIMA(p, d, q)** model — autoregressive integrated moving-average — where the middle parameter d is the number of times the series must be differenced to achieve stationarity. The ARIMA framework thus unifies the whole procedure: difference the series d times to make it stationary, then fit an ARMA model with p autoregressive and q moving-average terms to the result. Choosing the orders (p, d, q) is guided by the ACF and PACF and by the information criteria (AIC, BIC) of Chapter 11, a procedure associated with the names Box and Jenkins, and the fitted model is validated by checking that its residuals are white noise.

15.6 Forecasting

The payoff of a fitted time-series model is forecasting — projecting the series forward. A model produces not only a point forecast for each future period but, crucially, a *forecast interval* expressing the uncertainty around it, and the structure of that uncertainty is itself informative: forecast intervals widen as the horizon lengthens, because uncertainty compounds the further ahead one projects, and for nonstationary series they widen without bound. This is the time-series face of the prediction-interval idea from Chapter 10, with the added feature that uncertainty accumulates over time. The honest reporting of forecast uncertainty, and its growth with the horizon, is as important as the point forecast itself; a forecast without an interval conveys a false precision, and the widening of the interval is the model's candid admission of how quickly the predictable structure is exhausted.

15.7 Summary

Time series differ fundamentally from the independent samples of earlier chapters in that their ordering encodes dependence between observations — the very dependence that makes forecasting possible. The governing concept is stationarity, the constancy of statistical properties over time, which most real series achieve only after differencing to remove trends and stabilize variance. The dependence structure of a stationary series is summarized by its autocorrelation and partial autocorrelation functions, with white noise as the benchmark of pure unpredictability. The random walk exemplifies the nonstationary unit-root case, whose detection matters because regressing nonstationary series on one another produces spurious relationships. Stationary dependence is modeled by autoregressive and moving-average terms, combined in ARMA and, with differencing, ARIMA models, identified through the ACF, PACF, and information criteria and validated by white-noise residuals. The reward is forecasting with honestly widening intervals. This chapter also closes the loop on the autocorrelation diagnostics of Chapter 12, showing that what was there a violation to be corrected is here the structure to be modeled.

Chapter 16

Logistic Regression and Generalized Linear Models

The regression of Part IV models a continuous outcome as a linear function of predictors. A great many outcomes are not continuous: they are binary (an event occurs or does not), or counts (how many times something happens). Forcing linear regression onto such outcomes fails in specific ways, and the remedy is a broader framework — the **generalized linear model (GLM)** — that keeps the linear-predictor machinery while adapting it to the nature of the outcome. The most important special case, **logistic regression** for binary outcomes, is among the most widely used models in all of applied statistics, and it draws together the Bernoulli distribution of Chapter 5 and the maximum likelihood estimation of Chapter 7.

16.1 Why linear regression fails for binary outcomes

Suppose the outcome is binary, coded 0 or 1, and we want to model the probability p that it equals 1 as a function of predictors. Fitting an ordinary linear regression to the $0/1$ outcome — a "linear probability model" — runs into three concrete problems. First, the fitted line is unbounded, so it produces predicted probabilities below zero and above one, which are meaningless. Second, the errors of a binary outcome cannot be normal — they take only two values for each observation — violating the normality assumption. Third, the variance of a Bernoulli outcome is $p(1-p)$ (Chapter 5), which depends on p and hence on the predictors, so the errors are inherently heteroskedastic (Chapter 12). The linear model is simply the wrong shape for a probability, and what is needed is a model that maps the unbounded linear predictor into the $[0, 1]$ interval where probabilities live.

16.2 Odds, log-odds, and the logistic function

The bridge from probabilities to an unbounded scale runs through the *odds*. The **odds** of an event are the ratio of the probability it happens to the probability it does not, $p/(1-p)$ — a quantity ranging from zero to infinity as p ranges from zero to one. Taking the logarithm of the odds gives the **log-odds** or **logit**,

$$\text{logit}(p) = \ln \frac{p}{1-p},$$

which ranges over the entire real line: it goes to $-\infty$ as $p \to 0$, equals zero at $p = 1/2$, and goes to $+\infty$ as $p \to 1$. The logit has thus stretched the bounded probability scale onto the unbounded real line, exactly the scale on which a linear predictor lives. Inverting the logit gives the **logistic** (or sigmoid) function, which squashes any real number back into $(0, 1)$:

$$p = \frac{1}{1 + e^{-z}}, \qquad z = \text{logit}(p).$$

This S-shaped function is the heart of logistic regression: it takes an unbounded linear combination of predictors and returns a valid probability, flattening out gracefully as it approaches zero and one so that predictions never escape the legitimate range.

16.3 The logistic regression model

Logistic regression models the log-odds of the outcome as a linear function of the predictors:

$$\ln\frac{p_i}{1-p_i} = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip}.$$

Equivalently, the probability itself is the logistic function applied to the linear predictor. The model preserves everything attractive about linear regression — a linear combination of predictors, one coefficient per predictor, the partial-effect interpretation — while guaranteeing valid probabilities through the logit link.

The coefficients are interpreted on the log-odds scale, and translating them is essential. A one-unit increase in predictor X_j adds β_j to the log-odds, holding the other predictors fixed; exponentiating, it *multiplies the odds* by e^{β_j}. The quantity e^{β_j} is therefore the **odds ratio** for that predictor, and it is the natural way to report logistic results: an odds ratio of 2 means the odds of the outcome double for each unit increase in the predictor, an odds ratio of 1 means no effect, and one below 1 means the predictor lowers the odds. The effect on the *probability* is not constant — because of the S-shape, the same change in log-odds moves the probability a lot near $p = 1/2$ and little near the extremes — which is why effects are reported as odds ratios rather than as a single "change in probability."

16.4 Estimation by maximum likelihood

Logistic regression cannot be fit by least squares; there is no normal-error linear structure to minimize. Instead it is fit by **maximum likelihood**, the principle of Chapter 7: choose the coefficients that make the observed pattern of zeros and ones most probable. The likelihood is built from the Bernoulli probability of each observed outcome under the model, and the log-likelihood is maximized to find the estimates. Unlike the least-squares formula of Chapter 11, this maximization has no closed-form solution and must be carried out numerically, by iterative algorithms that converge to the maximum. The payoff is that the maximum likelihood machinery brings its general large-sample guarantees from Chapter 7: the estimates are consistent and asymptotically normal, which is what allows the coefficients to be tested and given confidence intervals in the familiar way, now using the asymptotic normality of the maximum likelihood estimator rather than exact t-distributions.

16.5 The generalized linear model framework

Logistic regression is one instance of a general and elegant template. A **generalized linear model** has three components: a *random component*, the distribution assumed for the outcome (Bernoulli for logistic regression, normal for ordinary regression, Poisson for counts); a *systematic component*, the linear predictor $\beta_0 + \beta_1 X_1 + \cdots$ that combines the predictors exactly as in ordinary regression; and a *link function*, which connects the mean of the outcome to the linear predictor. Ordinary linear regression is the special case with a normal random component and an identity link (the mean equals the linear predictor directly); logistic regression uses a Bernoulli random component and the logit link. This unifying view shows that the linear-predictor idea extends far beyond continuous outcomes — one chooses the distribution to match the data and the link to map its mean onto the unbounded linear scale, and the same estimation and inference principles apply throughout.

The other widely used member of the family is **Poisson regression**, for count outcomes (Chapter 5). It pairs a Poisson random component with a log link, modeling the log of the expected count as a linear function of the predictors, so that predicted counts are guaranteed

positive and coefficients are interpreted as multiplicative effects on the rate. Poisson regression is the natural model for outcomes that are counts of events — incidents, arrivals, occurrences — and it shares the maximum likelihood estimation and GLM structure of logistic regression. When count data are overdispersed (variance exceeding the mean, violating the Poisson's equality of the two, as flagged in Chapter 5), extensions such as negative binomial regression are used instead.

16.6 Assessing fit and classification

Because logistic and other GLMs are fit by likelihood rather than least squares, the goodness-of-fit measures differ from R^2. The central quantity is the **deviance**, a likelihood-based measure of lack of fit that generalizes the residual sum of squares, and nested models are compared by *likelihood ratio tests* — the GLM counterpart of the nested F-test of Chapter 11 — which check whether adding predictors significantly improves the likelihood. Various "pseudo-R^2" measures attempt to convey explained variation on a familiar scale, though none has quite the clean interpretation of the ordinary R^2.

When logistic regression is used for **classification** — predicting which class an observation belongs to by thresholding the predicted probability — its performance is assessed by how well the predicted classes match reality. This is summarized in a *confusion matrix* cross-tabulating predicted against actual classes, from which measures such as accuracy, sensitivity (the true-positive rate), and specificity (the true-negative rate) are computed. Because the classification threshold is a free choice that trades sensitivity against specificity, the *ROC curve*, which traces this tradeoff across all thresholds, and the area under it provide a threshold-independent summary of classification performance. These ideas connect logistic regression to the broader enterprise of statistical classification and machine learning, where it serves as both a workhorse model and the conceptual foundation for more elaborate methods.

16.7 Summary

When outcomes are binary or counts rather than continuous, ordinary linear regression is the wrong shape, producing impossible predictions and violating its error assumptions. The generalized linear model framework resolves this by keeping the linear predictor while choosing a distribution to match the outcome and a link function to map its mean onto the unbounded linear scale. Logistic regression, its most important case, models the log-odds of a binary outcome linearly via the logit link, with coefficients interpreted as odds ratios through exponentiation, and is fit by maximum likelihood with all the asymptotic inference that entails. Poisson regression handles counts through a log link in the same way. Fit is assessed through deviance and likelihood ratio tests, and when used for classification, logistic regression is evaluated through the confusion matrix and ROC curve. The framework extends the reach of regression from continuous measurements to the binary and count outcomes that pervade applied work, all within a single unified structure resting on the distributions of Chapter 5 and the maximum likelihood of Chapter 7.

Chapter 17

Bayesian Inference

The inference of Part III is *frequentist*: it treats the unknown parameter as a fixed constant and the data as random, and it quantifies uncertainty through the behavior of procedures over hypothetical repeated samples — the meaning, recall, of a confidence interval. There is an alternative paradigm, with a different philosophy and a different and arguably more intuitive interpretation of its results: **Bayesian inference**. It treats the parameter itself as uncertain, describes that uncertainty with a probability distribution, and updates the distribution as data arrive, using exactly the Bayes' theorem introduced in Chapter 3. This chapter develops the Bayesian approach, which has grown from a philosophical position into an indispensable practical toolkit.

17.1 Two views of probability

The two paradigms rest on different conceptions of what probability *is*. The frequentist view holds that probability is a long-run frequency, an objective property of repeatable experiments; a parameter is a fixed number with no probability distribution, because it does not vary across repetitions. The Bayesian view holds that probability is a degree of belief, a quantified state of knowledge; a parameter, though fixed in the world, is *unknown*, and our uncertainty about it can and should be expressed as a probability distribution. This is not mere philosophy — it leads to genuinely different procedures and, importantly, to differently interpretable answers. The Bayesian treats the question "what do I believe about this parameter, given what I have seen?" as the natural object of inference, and answers it directly.

17.2 The mechanics of updating

Bayesian inference is the systematic revision of belief in light of evidence, and its engine is Bayes' theorem applied to parameters. Three objects appear.

Definition 17.1 (Prior, likelihood, posterior)**.** The **prior** distribution $p(\theta)$ expresses our belief about the parameter θ *before* seeing the data. The **likelihood** $p(\text{data} \mid \theta)$ is the probability of the observed data as a function of θ — the same likelihood that drove maximum likelihood estimation in Chapter 7. The **posterior** distribution $p(\theta \mid \text{data})$ expresses our updated belief *after* seeing the data.

Bayes' theorem connects them in the form that is the master equation of the whole subject:

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta)\, p(\theta)}{p(\text{data})} \;\propto\; p(\text{data} \mid \theta)\, p(\theta),$$

usually read as the proportionality on the right: the posterior is proportional to the likelihood times the prior. The denominator is a constant that does not depend on θ and serves only to make the posterior integrate to one, so the *shape* of the posterior is entirely determined by multiplying the likelihood by the prior. This single relationship is the whole of Bayesian inference in principle: start with a prior, multiply by the likelihood of the data, and obtain the

posterior, which is the complete answer — a full distribution over the parameter, not a single estimate. Every Bayesian quantity is then read off this posterior: its mean or mode is the point estimate, its spread is the uncertainty, and intervals of it are the interval estimates.

The relationship also makes the role of data transparent. With little data the likelihood is diffuse and the prior exerts substantial influence on the posterior; as data accumulate the likelihood sharpens and increasingly dominates, so that with enough data the posterior is driven by the evidence and the choice of prior washes out. This is the formal sense in which Bayesian inference learns: belief is dragged by evidence toward the truth, quickly when data are plentiful and gently when they are scarce, and two analysts who began with different reasonable priors will converge as the data mount.

17.3 The prior and its controversy

The distinctive ingredient is the prior, and it is the source of both the method's power and its long controversy. A prior may be *informative*, encoding genuine prior knowledge — previous studies, physical constraints, expert judgment — which is a strength when such knowledge exists and a Bayesian analysis can incorporate it where a frequentist one cannot. Or it may be *uninformative* (flat or diffuse), chosen to express ignorance and let the data speak, which keeps the analysis close to the likelihood and hence often close to frequentist results. The objection historically leveled at Bayesian inference is that the prior is *subjective*: two analysts may choose different priors and reach different conclusions from the same data. The Bayesian responses are twofold and both have force. First, the subjectivity is explicit and auditable — the prior is stated and can be examined and varied in a sensitivity analysis — whereas frequentist analyses smuggle in assumptions of their own that are no less consequential for being less visible. Second, as just noted, the influence of the prior vanishes as data accumulate, so for well-identified problems with adequate data the choice matters little. The honest practice is to report how conclusions depend on the prior when data are limited and the prior is doing real work.

17.4 Conjugate priors and a worked example

In general, multiplying a likelihood by a prior produces a posterior with no simple closed form, but for certain matched pairs of prior and likelihood the posterior belongs to the same family as the prior, making the update analytically clean. Such a prior is called **conjugate** to the likelihood. The most elegant example, and a genuinely useful one, is the Beta–Binomial pair for estimating an unknown probability.

Example 17.1 (Beta–Binomial updating)**.** Suppose we wish to estimate an unknown success probability θ — the chance an event occurs. The natural prior for a probability is a **Beta** distribution, $\text{Beta}(\alpha, \beta)$, defined on $[0, 1]$ and flexible enough to express anything from complete ignorance (the flat $\text{Beta}(1, 1)$, which is uniform on $[0, 1]$) to strong prior belief. The parameters α and β behave like prior counts of successes and failures. We then observe data: k successes in n independent trials, whose likelihood is Binomial (Chapter 5). Multiplying the Binomial likelihood by the Beta prior, the posterior is again a Beta distribution,

$$\theta \mid \text{data} \;\sim\; \text{Beta}(\alpha + k,\ \beta + n - k).$$

The update could not be simpler: add the observed successes to α and the observed failures to β. The posterior mean is $(\alpha + k)/(\alpha + \beta + n)$, which is a weighted blend of the prior mean and the observed success fraction k/n — and the weight on the data grows with n, so that for large samples the posterior mean approaches the observed fraction, recovering the frequentist

estimate, while for small samples the prior pulls the estimate toward its own mean, a built-in and principled form of shrinkage. Concretely, starting from the flat prior $\text{Beta}(1,1)$ and observing 7 successes in 10 trials gives a $\text{Beta}(8,4)$ posterior with mean $8/12 \approx 0.67$, slightly pulled from the raw 0.7 toward the prior mean of 0.5, with the entire posterior distribution available to describe the remaining uncertainty.

Other conjugate pairs play the same role in other settings — a Normal prior is conjugate to a Normal likelihood for estimating a mean, and a Gamma prior is conjugate to a Poisson likelihood for estimating a rate — and in each case the posterior is a tidy update of the prior's parameters by the data. Conjugacy is mathematically convenient but not required; it simply makes the arithmetic transparent in these foundational cases.

17.5 Credible intervals

Bayesian interval estimation produces a **credible interval**, and its interpretation is the one that the confidence interval of Chapter 7 conspicuously *lacked*.

Definition 17.2 (Credible interval)**.** A 95% **credible interval** is an interval containing 95% of the posterior probability for the parameter.

The crucial point is that a credible interval means exactly what people *wish* a confidence interval meant: given the data and the prior, there is a 95% probability that the parameter lies within it. This direct probability statement about the parameter is legitimate in the Bayesian framework precisely because the parameter is treated as a random variable with a posterior distribution, so it is meaningful to speak of the probability that it falls in a range. Recall that the frequentist confidence interval does *not* support this reading — the 95% there refers to the long-run coverage of the procedure, not to any single interval. The credible interval thus resolves the most common misinterpretation of confidence intervals by making the natural interpretation actually correct. For problems with abundant data and flat priors, credible and confidence intervals often nearly coincide numerically, but their meanings differ, and the Bayesian meaning is the more intuitive.

17.6 Computation and the modern resurgence

Bayesian inference was for a long time limited to the conjugate cases where the posterior could be computed by hand, because realistic models produce posteriors that are high-dimensional and analytically intractable. The obstacle was computational, and it was overcome by **Markov chain Monte Carlo (MCMC)**, a family of algorithms that, rather than computing the posterior in closed form, *draw samples* from it, building up an empirical picture of the posterior from which any quantity — means, intervals, probabilities — can be estimated. MCMC and related computational methods transformed Bayesian inference from a niche philosophical stance into a practical tool applicable to arbitrarily complex models, and they are the reason Bayesian methods now pervade modern statistics, machine learning, and the sciences. The conceptual content is unchanged — posterior proportional to likelihood times prior — but computation makes it usable far beyond the textbook conjugate examples.

17.7 Bayesian and frequentist, in practice

Neither paradigm is universally correct; they answer subtly different questions and each has natural domains. Bayesian methods shine when genuine prior information exists and should be incorporated, when a direct probability statement about a parameter is wanted, when data

are scarce so that principled shrinkage toward a sensible prior is valuable, and when complex hierarchical models are needed. Frequentist methods are natural when an objective, prior-free analysis is desired, when long-run error control is the explicit goal, and for the standard tests and intervals that are well understood and computationally trivial. In much routine work with ample data the two converge to similar numerical answers, and the choice is then one of interpretation and convenience. The mature view is not tribal: the two frameworks are complementary lenses, and the well-equipped analyst understands both, choosing according to the question, the available prior knowledge, and the interpretation required. What unites them is the subject of this entire book — the disciplined use of probability to reason from data under uncertainty — approached, in the Bayesian case, by turning the Bayes' theorem of Chapter 3 into a complete engine of inference.

17.8 Summary

Bayesian inference treats an unknown parameter as itself uncertain, described by a probability distribution that is updated by data through Bayes' theorem: the posterior is proportional to the likelihood times the prior. The prior encodes belief before the data — informative when knowledge exists, diffuse when it does not — and although its subjectivity has been controversial, it is explicit, auditable, and washed out by sufficient data. Conjugate priors such as the Beta for a probability make the update a transparent adjustment of parameters, as the Beta–Binomial example shows, with the posterior blending prior and data and shrinking toward the prior when data are scarce. The resulting credible interval carries the direct probability interpretation that the frequentist confidence interval cannot, resolving its most common misreading. Freed by Markov chain Monte Carlo from the limits of analytic tractability, Bayesian methods now apply to models of arbitrary complexity. Bayesian and frequentist inference are complementary, differing in philosophy and interpretation but united in their purpose, and the Bayesian framework is in a real sense the Bayes' theorem of Chapter 3 elevated into a complete and practical system of inference.

Chapter 18

Resampling and Modern Computational Methods

The inference of Part III derived sampling distributions analytically — the normal for a mean, the t when the variance is estimated, the chi-squared and F for variances and models. This works beautifully when the assumptions hold and the mathematics is tractable, but many statistics of practical interest have sampling distributions that are intractable to derive, and many real datasets violate the assumptions on which the analytic results depend. The modern response is to let computation substitute for mathematics: rather than deriving a sampling distribution, *simulate* it by resampling the data itself. This chapter presents the principal resampling methods — the bootstrap, permutation tests, and cross-validation — together with regularization, which operationalizes the bias–variance tradeoff that has recurred throughout the book. These techniques define much of contemporary statistical practice.

18.1 The bootstrap

The deepest idea in computational statistics is also one of the simplest to state. We want the sampling distribution of a statistic — how it varies across hypothetical repeated samples from the population — but we have only one sample and do not know the population. The **bootstrap** confronts this by using the one sample we have *as a stand-in* for the population, and resampling from it.

Definition 18.1 (The bootstrap)**.** The **bootstrap** approximates the sampling distribution of a statistic by repeatedly drawing new samples, each of the same size as the original, by sampling *with replacement* from the observed data, computing the statistic on each resample, and using the resulting collection of values as an estimate of the statistic's sampling distribution.

The logic rests on a single substitution. The true sampling distribution arises from drawing samples from the population; since we lack the population, we draw instead from the *empirical distribution* — the observed data, treated as if it were the population, each data point given equal probability. Sampling with replacement from the data mimics drawing fresh samples from a population that looks exactly like our sample, and because the sample is our best available picture of the population, the variability across these resamples approximates the variability across genuine repeated samples. Drawing with replacement is essential: it allows each resample to differ from the original by including some observations multiple times and omitting others, which is what generates the variation that estimates the sampling distribution.

From the bootstrap distribution one reads off everything the analytic sampling distribution would have provided. The standard deviation of the bootstrapped statistic values is a **bootstrap standard error**, and the percentiles of the bootstrap distribution give a **bootstrap confidence interval** — for a 95% interval, the range between the 2.5th and 97.5th percentiles of the resampled values. The power of the method is its generality: it requires no formula for the standard error and no assumption of normality, so it applies to statistics for which deriving a sampling distribution would be hopeless — a median, a correlation, a ratio, a complicated estimator with no analytic theory. Wherever the classical approach demands

a tractable formula and convenient assumptions, the bootstrap demands only the ability to recompute the statistic many times, trading mathematical derivation for computation. Its main limitations are that it still assumes the original sample is representative of the population — it cannot repair a biased sample — and that it can be unreliable for statistics that depend on extreme tail behavior, such as the maximum, where a single sample contains too little information about the tail.

18.2 Permutation tests

The bootstrap estimates a sampling distribution; for hypothesis testing, a closely related resampling idea constructs a *null* distribution directly. A **permutation test** assesses whether two groups differ by asking what the data would look like if the group labels were meaningless. If, under the null hypothesis, the labels carry no information, then any reassignment of labels to observations is as likely as the one actually observed. The test therefore repeatedly *shuffles* the labels, recomputing the statistic of interest — a difference in means, say — for each shuffle, and builds up the distribution of the statistic under the null hypothesis that the labels do not matter. The observed statistic is then compared to this permutation distribution, and its rank within that distribution yields the p-value: if the genuine labeling produces a statistic far out in the tail of what random relabelings produce, the labels evidently do matter, and the null is rejected.

Permutation tests have two attractive features. They are essentially *assumption-free*, requiring no distributional model — the null distribution is constructed from the data themselves — which makes them a natural partner to the nonparametric methods of Chapter 14. And they are *exact* in the sense that the null distribution is built directly rather than approximated by an asymptotic formula, so their validity does not rely on large samples. They are, in effect, the computational realization of the same distribution-free philosophy that motivated rank tests, achieved by reshuffling rather than ranking, and they extend naturally to settings where no classical test exists.

18.3 Cross-validation

Chapter 11 argued that the honest measure of a model is its performance on data it was not fit to, because a model selected to maximize in-sample fit flatters itself by capturing noise. **Cross-validation** is the resampling method that operationalizes this principle, turning out-of-sample evaluation into a systematic procedure. The simplest form holds out a portion of the data as a test set, fits the model on the rest, and measures error on the held-out portion — but this wastes data and depends on the luck of the split. ***k*-fold cross-validation** resolves both problems: the data are divided into k equal parts, or folds; the model is fit k times, each time leaving out one fold and testing on it; and the k test errors are averaged into a single estimate of out-of-sample performance. Every observation is used for testing exactly once and for training the other $k - 1$ times, extracting honest validation from limited data.

Cross-validation is the standard tool for two tasks. It *estimates* how well a model will generalize, giving a realistic forecast of performance on new data rather than the optimistic in-sample figure. And it *selects* among models or tunes their settings — comparing candidate models or values of a tuning parameter by their cross-validated error and choosing the best — which is a more reliable basis for model selection than the in-sample criteria of Chapter 11 precisely because it directly targets out-of-sample performance. It is the practical embodiment of the lesson that genuine validation requires data not used in fitting, and it is ubiquitous in modern predictive modeling.

18.4 Regularization

The final method addresses overfitting not by evaluating models after the fact but by changing how they are fit, and it is the operational face of the bias–variance tradeoff introduced in Chapter 7. Ordinary least squares minimizes only the residual sum of squares, and when predictors are numerous or collinear (Chapter 11) this yields unstable, high-variance coefficients that fit the sample's noise. **Regularization** adds a *penalty* on the size of the coefficients to the least-squares objective, deliberately introducing a little bias in exchange for a large reduction in variance, in the hope of lowering the overall mean squared error per the decomposition of Chapter 7.

Two penalties dominate. **Ridge regression** adds a penalty proportional to the sum of the *squared* coefficients, shrinking them all toward zero — and shrinking the coefficients of collinear predictors especially, which stabilizes the wild estimates that multicollinearity produces without discarding any predictor. **Lasso regression** adds a penalty proportional to the sum of the *absolute* coefficients, which has the distinctive and valuable property of driving some coefficients exactly to zero, thereby performing *variable selection* automatically: the lasso simultaneously fits and prunes the model, yielding a sparse solution that retains only the predictors that earn their place. Both methods are governed by a tuning parameter controlling the strength of the penalty — from none (recovering ordinary least squares) to severe (shrinking everything toward zero) — and that parameter is chosen, fittingly, by cross-validation. Regularization is one of the central techniques of modern statistics and machine learning, and it is best understood not as a trick but as the bias–variance tradeoff made into a fitting procedure: accept some bias from shrinkage to buy a large variance reduction, and let out-of-sample validation set the balance.

18.5 The computational turn

The methods of this chapter share a common spirit that marks a genuine shift in how statistics is practiced. Where the classical theory of Part III relied on deriving sampling distributions analytically under convenient assumptions, the computational approach *simulates* what it needs — resampling the data to estimate a sampling distribution (the bootstrap), to construct a null distribution (permutation tests), or to evaluate out-of-sample performance (cross-validation) — and reshapes the fitting problem to control complexity directly (regularization). The trade is assumptions for computation: these methods ask less of the data's conformity to mathematical ideals and more of the processor, a bargain that modern computing makes overwhelmingly favorable. They do not replace the analytic theory, which remains the foundation that explains *why* they work and provides the benchmarks they are checked against, but they extend its reach to the many problems where the classical formulas run out — complicated statistics, violated assumptions, and the high-dimensional predictive models that define contemporary data analysis.

18.6 Summary

Resampling methods substitute computation for analytic derivation, simulating the quantities that classical theory computes by formula. The bootstrap approximates a statistic's sampling distribution by resampling the data with replacement, yielding standard errors and confidence intervals for statistics with no tractable theory and without assuming normality. Permutation tests construct a null distribution directly by shuffling labels, providing exact, assumption-free hypothesis tests in the spirit of the nonparametric methods of Chapter 14. Cross-validation operationalizes the principle that models must be judged out of sample, using k-fold splitting

to estimate generalization and to select models honestly. Regularization — ridge and lasso — confronts overfitting at the fitting stage by penalizing coefficient size, realizing the bias–variance tradeoff of Chapter 7 as a procedure and, in the lasso's case, performing automatic variable selection, with cross-validation tuning the penalty. Together these techniques trade mathematical assumptions for computational effort, a bargain that defines modern statistical practice and that rests, throughout, on the conceptual foundations the rest of this book has built.